Teichmüller Coordinates for Nondegenerate Polygonable One-Holed Dilation Torus

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Abstract

Let S be a one-holed torus with a marked boundary point, and let $\mathcal{T}^{\dagger}_{\mathrm{poly}}(S)$ denote the marked Teichmüller space of polygonable dilation structures with nontrivial linear holonomy and nonzero boundary translation. We prove that the enhanced holonomy map

$$\widehat{\mathrm{Hol}} : \mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \longrightarrow \mathcal{E}^{\dagger}$$

is a regular $\mathbb{Z}$-covering. Moreover,

$$\mathcal{E}^{\dagger} \cong \mathcal{B} \times (\mathbb{R}^2 \setminus \{0\}), \qquad \mathcal{B} \cong SL_2(\mathbb{R}),$$

and this covering is precisely the universal covering in the second factor. Consequently,

$$\mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \cong \mathcal{B} \times \mathbb{R}^2 \cong SL_2(\mathbb{R}) \times \mathbb{R}^2.$$

1 Introduction and main theorem

Theorem 1.1 (Global Teichmüller parametrization)**.** *There are real-analytic identifications*

$$\mathcal{E}^\dagger \cong \mathcal{B} \times (\mathbb{R}^2 \setminus \{0\}), \qquad \mathcal{B} \cong SL_2(\mathbb{R}),$$

under which

$$\widehat{\mathrm{Hol}} : \mathcal{T}^\dagger_{\mathrm{poly}}(S) \longrightarrow \mathcal{E}^\dagger$$

is a connected regular $\mathbb{Z}$-covering. More precisely, there is a real-analytic diffeomorphism

$$\Psi : \mathcal{T}^\dagger_{\mathrm{poly}}(S) \xrightarrow{\sim} \mathcal{B} \times \mathbb{R}^2$$

such that

$$(\mathrm{id}_{\mathcal{B}} \times \varpi) \circ \Psi = \widehat{\mathrm{Hol}},$$

where

$$\varpi(s, \theta) = e^s \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}.$$

Consequently,

$$\mathcal{T}^\dagger_{\mathrm{poly}}(S) \cong SL_2(\mathbb{R}) \times \mathbb{R}^2.$$

The deck group is $\mathbb{Z}$, acting by

$$(\beta, s, \theta) \longmapsto (\beta, s, \theta + 2\pi k), \qquad k \in \mathbb{Z}.$$

2 Notation and main objects

Let S be an oriented compact genus-one surface with one boundary component and a distinguished point $s \in \partial S$.

A *dilation structure* on S is an atlas on $S \setminus \{s\}$ with dilation transitions. The structure is *polygonable* if it arises from a pentagon.

A marked structure is a pair (X, f), with $f : S \to X$ orientation preserving and $f(s)$ the singularity.

Two pairs are equivalent if a dilation isomorphism $h : X \to X'$, locally $z \mapsto rz + c$ with $r > 0$, makes $f'^{-1}hf$ isotopic to the identity through homeomorphisms preserving the boundary setwise and fixing s.

Let $\mathcal{T}_{\mathrm{Dil}}(S)$ be this marked deformation space.

The subspace $\mathcal{T}^{\dagger}_{\mathrm{poly}}(S)$ consists of polygonable structures with nontrivial linear holonomy and nonzero boundary translation .

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2.1 Room space

Define the **algebraic room space** by $\mathcal{Y}_{\mathrm{alg}} := (\mathbb{C}^2)/\mathbb{R}_{>0} \times \mathbb{R}^2$. Define the **basis space** by

$$\mathcal{B} := \left\{(e_1, e_2) \in \mathbb{C}^2/\mathbb{R}_{>0} \;\middle|\; \omega(e_1, e_2) > 0\right\} \cong SL_2(\mathbb{R}).$$

The **strict convex room space** and the **room space** are

$$\mathcal{Y}_{++} := \mathcal{B} \times \mathbb{R}^2_{>0} \subset \mathcal{Y} := \mathcal{B} \times \mathbb{R}^2 \setminus \{0\} = \{[e_1, e_2; \ell_1, \ell_2] \in \mathcal{Y}_{\mathrm{alg}} \mid [e_1, e_2] \in \mathcal{B},\ (\ell_1, \ell_2) \neq \{0\}\}.$$

and the **weak convex room space** as $\mathcal{Y}_+ := \mathcal{B} \times \left(\mathbb{R}^2_{\geq 0} \setminus \{(0,0)\}\right) \setminus \mathcal{Y}_{++}$. Then for a element

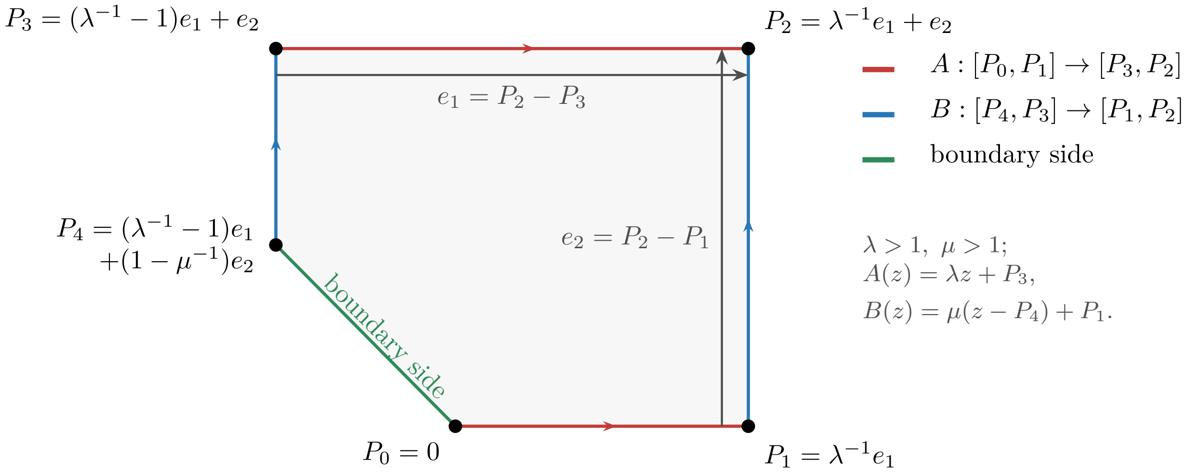
$$q = [e_1, e_2; \ell_1, \ell_2] \in \mathcal{Y}_{\mathrm{alg}}, \qquad \lambda = e^{\ell_1}, \quad \mu = e^{\ell_2}.$$

the algebraic room is defined by

$$\begin{aligned} &P_0 = 0, && P_1 = \lambda^{-1} e_1, && P_2 = \lambda^{-1} e_1 + e_2, \\ &P_3 = (\lambda^{-1} - 1)e_1 + e_2, && P_4 = (\lambda^{-1} - 1)e_1 + (1 - \mu^{-1})e_2, && \end{aligned} \tag{1}$$

where $[P_4, P_0]$ is the boundary side and side paring dilation maps

$$A(z) = \lambda z + P_3 : [P_0, P_1] \to [P_3, P_2], \quad B(z) = \mu z + (P_1 - \mu P_4) : [P_4, P_3] \to [P_1, P_2]. \tag{2}$$



For a pentagon room $q \in \mathcal{Y}$, from Eq. (1), we have

$$\begin{aligned} &P_1 - P_0 = \lambda^{-1} e_1, \qquad P_2 - P_1 = e_2, \qquad P_3 - P_2 = -e_1, \\ &P_4 - P_3 = -\mu^{-1} e_2, \qquad P_0 - P_4 = (1 - \lambda^{-1})e_1 - (1 - \mu^{-1})e_2. \end{aligned} \tag{3}$$

A room is **geometrically convex in Euclidean space** if all vertex turnings have the same sign, which is determined by the determinants of consecutive oriented edge vectors:

$$\begin{aligned} &\omega(P_1 - P_0, P_2 - P_1) = \lambda^{-1} \omega(e_1, e_2), && \omega(P_2 - P_1, P_3 - P_2) = \omega(e_1, e_2), \\ &\omega(P_3 - P_2, P_4 - P_3) = \mu^{-1} \omega(e_1, e_2), && \omega(P_4 - P_3, P_0 - P_4) = \mu^{-1}(1 - \lambda^{-1})\omega(e_1, e_2), \\ &\omega(P_0 - P_4, P_1 - P_0) = \lambda^{-1}(1 - \mu^{-1})\omega(e_1, e_2). && \end{aligned} \tag{4}$$

So we say a room $q \in \mathcal{Y}$ is **strict convex** if and only if

$$q \in \mathcal{Y}_{++} \iff \lambda > 1,\ \mu > 1 \iff \ell_1 > 0,\ \ell_2 > 0, \tag{5}$$

and room $q \in \mathcal{Y}$ is **weak convex** if and only if

$$q \in \mathcal{Y}_+ \iff (\lambda = 1,\ \mu > 1) \text{ or } (\lambda > 1,\ \mu = 1) \iff (\ell_1 = 0,\ \ell_2 > 0) \text{ or } (\ell_1 > 0,\ \ell_2 = 0). \tag{6}$$

2.2 Enhanced holonomy space

For $\pi_1(S) = F_2 = \langle a, b\rangle$, write

$$A := \rho(a) = (\lambda, u_0) : z \to \lambda z + u_0, \qquad B := \rho(b) = (\mu, v_0) : z \to \mu z + v_0, \qquad \lambda, \mu > 0. \tag{7}$$

The boundary word for the room $q \in \mathcal{Y}$ is

$$\partial_0 := a^{-1}b^{-1}ab = (ba)^{-1}[a, b](ba), \qquad \rho(\partial_0)(P_4) = P_0, \qquad [a, b] := aba^{-1}b^{-1}. \tag{8}$$

Writing translation map as $T_w(z) := z + w$, one has

$$\rho([a, b]) = T_\tau, \qquad \rho(\partial_0) = T_\delta, \qquad \tau = (1-\mu)u_0 + (\lambda-1)v_0, \qquad \delta = \frac{\tau}{\lambda\mu}. \tag{9}$$

Fix a universal lift $\widetilde{s}_0 \in \pi^{-1}(s)$ for singularity $s \in \partial S$ and set $p = \widehat{\mathrm{dev}}(\widetilde{s}_0)$, **enhanced holonomy space** is defined by

$$\mathcal{E} := \{(\rho, p) \mid \rho : F_2 \to \mathrm{Dil}^+(\mathbb{C}),\ p \in \mathbb{C}\}/\,\mathrm{Dil}^+(\mathbb{C}), \quad g(\rho, p) = (g\rho g^{-1}, g(p)) \tag{10}$$

Every element $[\rho, p] \in \mathcal{E}$ is determined by $[\lambda, \mu; u_0, v_0; p]$, and it is clear that we always have **normalized representation** $[\rho, p] \sim [T_{-p} \circ \rho \circ T_{-p}^{-1}, 0]$, which is given by $[\lambda, \mu; u_0, v_0; p] \sim [\lambda, \mu; (A-I)p, (B-I)p; 0]$. Put

$$U = A(p) - p = \widehat{\mathrm{dev}}\,(a \cdot \widetilde{s}_0) - \widehat{\mathrm{dev}}\,(\widetilde{s}_0)\,, \quad V = B(p) - p = \rho(b) \cdot p - p, \tag{11}$$

Then the normalized enhanced holonomy representation uniquely determine by

$$[\lambda, \mu; U, V; 0] \longleftrightarrow q = [e_1, e_2; \log\lambda, \log\mu] \in \mathcal{Y}_{\mathrm{alg}} = (\mathbb{C}^2)/\mathbb{R}_{>0} \times \mathbb{R}^2. \tag{12}$$

If we have a room date $q = [e_1, e_2; \log\lambda, \log\mu]$, by equation 2 we have $A(z) = \lambda z + P_3$ and $B(z) = \mu(z - P_4) + P_1$ and $p = P_0$, then

$$\begin{aligned} U &= P_3 = (\lambda^{-1} - 1)e_1 + e_2, \\ V &= P_1 - \mu P_4 = \left[1 + \left(1 - \lambda^{-1}\right)(\mu - 1)\right] e_1 + (1-\mu)e_2. \end{aligned} \tag{13}$$

Conversely, using Eq. (1), it suffices to set

$$e_1 = V + (\mu-1)U, \qquad e_2 = \left[1 + \left(1-\lambda^{-1}\right)(\mu-1)\right] U + \left(1 - \lambda^{-1}\right) V. \tag{14}$$

Since

$$\omega(e_1, e_2) = \omega\Big(e_1, U + (1-\lambda^{-1})e_1\Big) = \omega(e_1, U) = \omega(V + (\mu-1)U, U) = \omega(V, U) = -\omega(U, V). \tag{15}$$

we have the **nondegenerate enhanced holonomy space**

$$\mathcal{E}^\dagger := \{[(\rho, p)] \in \mathcal{E} \mid \ell(\rho) \neq 0, \omega(U, V) < 0\} = \{[e_1, e_2; \ell_1, \ell_2] \in \mathcal{Y}_{\mathrm{alg}} \ \mid \ell \neq 0, \omega(e_1, e_2) > 0\} = \mathcal{Y} \tag{16}$$

and

$$\mathcal{E}_{++} := \Big\{\eta \in \mathcal{E}^\dagger \mid \ell_1(\eta) > 0, \ell_2(\eta) > 0\Big\} = \{q \in \mathcal{Y} \mid \ell_1 > 0, \ell_2 > 0\} = \mathcal{Y}_{++}. \tag{17}$$

We have diffeomorphism between room space and nondegenerate enhanced holonomy space

$$\Phi : \mathcal{Y} \longrightarrow \mathcal{E}^\dagger, \qquad \Phi(q) = [(\rho, 0)], \quad \rho(a) = A, \quad \rho(b) = B, \quad \mathcal{E}_{++} := \Phi(\mathcal{Y}_{++}) \subset \mathcal{E}^\dagger. \tag{18}$$

And the enhanced holonomy map is a local homeomorphism

$$\widehat{\mathrm{Hol}} : \mathcal{T}^\dagger_{\mathrm{poly}}(S) \longrightarrow \mathcal{E}^\dagger, \qquad [X, f] \longmapsto [(\rho, \widehat{\mathrm{dev}}(\widetilde{s}_0))]. \tag{19}$$

2.3 The enhanced holonomy space is $\mathcal{X}^{\dagger}_{\mathrm{Dil}}(S) \times \mathbb{H}$

Denote $\ell(\rho) := \begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} = \begin{pmatrix} \log \lambda \\ \log \mu \end{pmatrix} \in \mathbb{R}^2$. The **ordinary nondegenerate character variety** is

$$\mathcal{X}^{\dagger}_{\mathrm{Dil}}(S) := \left\{ \rho \in \mathrm{Hom}(F_2, \mathrm{Dil}^+(\mathbb{C})) \;\middle|\; \ell(\rho) \neq 0,\ \tau \neq 0 \right\} / \, \mathrm{Dil}^+(\mathbb{C}), \qquad g \cdot \rho = g \rho g^{-1}.$$

and we have paramatrisation $(\ell, [\delta]) \in \mathcal{X}^{\dagger}_{\mathrm{Dil}}(S) \cong \mathbb{R}^2 \setminus \{0\} \times (\mathbb{C}^*/\mathbb{R}_{>0}), \delta := \frac{\tau}{\lambda\mu}$.

Proposition 2.1 (Product coordinates for the enhanced holonomy space)**.** *Using the coordinates $[\rho] \longleftrightarrow (\ell, [\delta])$ on $\mathcal{X}^{\dagger}_{\mathrm{Dil}}(S)$, we have diffeomorphism*

$$\Xi : \mathcal{Y} \cong \mathcal{E}^{\dagger} \longrightarrow \mathcal{X}^{\dagger}_{\mathrm{Dil}}(S) \times \mathbb{H}, \qquad \Xi(\Phi([e_1, e_2; \ell_1, \ell_2])) \longmapsto (\ell, [\delta], \frac{e_2}{e_1}), \qquad z := \frac{e_2}{e_1}.$$

Proof. **Well-definedness.** The conditions $\ell(\rho) \neq 0, \omega(U, V) < 0$ imply $\tau \neq 0$ and so $\delta \neq 0$: if $\tau = (1 - \mu)U + (\lambda - 1)V = 0$, then U, V are real-linearly independent, so we have $\lambda = \mu = 1$, contradicting $\ell(\rho) \neq 0$.

The condition $\omega(e_1, e_2) > 0$ is equivalent to $z \in \mathbb{H}$: let $q = [e_1, e_2; \ell_1, \ell_2] \in \mathcal{Y}$, and set $z := \frac{e_2}{e_1}$, then we have

$$\omega\left(e_1, e_2\right) = \mathrm{Im}\left(\overline{e_1} e_2\right) = \mathrm{Im}\left(\overline{e_1} e_1 \frac{e_2}{e_1}\right) = \mathrm{Im}\left(\left|e_1\right|^2 z\right) = \left|e_1\right|^2 \mathrm{Im}\, z.$$

To show it is a bijection we use the notation as

$$\delta = (1 - e^{-\ell_1}) e_1 + (e^{-\ell_2} - 1) e_2 = c_\ell(z) e_1, \qquad c_\ell(z) := 1 - e^{-\ell_1} + (e^{-\ell_2} - 1) z.$$

Then for every $\ell \neq 0$ and $z \in \mathbb{H}$, we have $c_\ell(z) \neq 0$: if $c_\ell(z) = 0$, then, since $\mathrm{Im}\, z > 0$, and $\mathrm{Im}\, c_\ell(z) = (e^{-\ell_2} - 1)\, \mathrm{Im}\, z$, we have $e^{-\ell_2} - 1 = 0$, hence we get $\mathrm{Re}\, c_\ell(z) = 1 - e^{-\ell_1} = 0$, so $\ell_1 = \ell_2 = 0$, contradicting $\ell \neq 0$.

Injectivity. Let $q = [e_1, e_2; \ell_1, \ell_2]$ and $q' = [e'_1, e'_2; \ell'_1, \ell'_2]$. Suppose

$$\left(\ell, [\delta], \frac{e_2}{e_1}\right) = \Xi(\Phi(q)) = \Xi(\Phi(q')) = \left(\ell', [\delta'], \frac{e'_2}{e'_1}\right).$$

Then since $\ell = \ell'$, we only need to show $[e'_1, e'_2] = [e_1, e_2] \in \mathcal{B}$.

Since $[\delta] = [\delta']$, there exists $r > 0$ such that $\delta' = r\delta$. On the other hand,

$$\delta = c_\ell(z) e_1, \qquad \delta' = c_\ell(z) e'_1.$$

Hence $e'_1 = r e_1$. Since $\frac{e_2}{e_1} = \frac{e'_2}{e'_1}$, we then get $e'_2 = r e_2$ too.

Surjectivity. Let $(\ell, [\delta]) \in \mathcal{X}^{\dagger}_{\mathrm{Dil}}(S)$ and $z \in \mathbb{H}$, choose $\delta \in [\delta]$ and set

$$e_1 := \frac{\delta}{c_\ell(z)}, \qquad e_2 := z e_1.$$

Since $c_\ell(z) \neq 0$, then this is well defined, and $\omega(e_1, e_2) = |e_1|^2\, \mathrm{Im}\, z > 0$. Hence we get

$$q = [e_1, e_2; \ell_1, \ell_2] \in \mathcal{Y}.$$

then q is independent of the choice of representative of $[\delta]$: If $\delta' = r\delta$ for $r > 0$, then $e'_1 = r e_1$ and $e'_2 = r e_2$.

□

3 Relative Mapping class group actions

Since the cut-and-paste operations leave the boundary side fixed pointwise, the relevant action should be induced by the relative mapping class group of the one-holed torus where the homeomorphisms fix ∂S pointwise,

$$\Gamma_\partial := \pi_0 \operatorname{Homeo}^+(S; \partial S), \tag{20}$$

Consider the identification

$$\iota : B_3 = \langle \sigma_1, \sigma_2 \mid \sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2 \rangle \xrightarrow{\cong} \Gamma_\partial, \tag{21}$$

given by (D_a, D_b denote Dehn twists along a, b.)

$$\iota(\sigma_1) = D_a^{-1}, \quad \iota(\sigma_2) = D_b^{-1}. \tag{22}$$

Since every element of fixes the base point $s \in \partial S$, it induces an automorphism of $F_2 = \pi_1(S, s)$. Thus we obtain a homomorphism

$$j : B_3 \longrightarrow \operatorname{Aut}(F_2), \tag{23}$$

given on the generators by

$$j(\sigma_1)(a) = a, \quad j(\sigma_1)(b) = a^{-1}b, \quad j(\sigma_2)(a) = ba, \quad j(\sigma_2)(b) = b. \tag{24}$$

Then **action on $\mathcal{T}^\dagger_{\mathrm{poly}}(S)$ is given by**

$$\gamma \cdot [X, f] = \left[X, f \circ \iota(\gamma)^{-1}\right]. \tag{25}$$

the developing pair become $[\operatorname{dev} \circ \widetilde{\iota(\gamma)}^{-1}, \ \rho \circ j(\gamma)^{-1}]$, since $\iota(\gamma)(s) = s$ we can choose $\widetilde{\iota(\gamma)}$ fixes $\widetilde{s}_0$, so the developed singularity still be p, consequently

Action on $\mathcal{E}^\dagger$ is given by:

$$\gamma \cdot [(\rho, p)] := \left[\left(\rho \circ j(\gamma)^{-1}, p\right)\right]. \tag{26}$$

More explicitly, if $\widehat{\operatorname{Hol}}([X, f]) = [(\rho, p)]$, then

$$\widehat{\operatorname{Hol}}(\gamma \cdot [X, f]) = \widehat{\operatorname{Hol}}([X, f \circ \iota(\gamma)^{-1}]) = \left[\left(\rho \circ j(\gamma)^{-1}, p\right)\right] = \gamma \cdot [(\rho, p)] = \gamma \cdot \widehat{\operatorname{Hol}}([X, f]). \tag{27}$$

Action on $\mathcal{Y}$ is given by

$$\gamma \cdot q := \Phi^{-1}(\gamma \cdot \Phi(q)) = \Phi^{-1}(\gamma \cdot [(\rho, 0)]) = \Phi^{-1}\left(\left[\left(\rho \circ j(\gamma)^{-1}, 0\right)\right]\right), \qquad \rho(a) = A, \quad \rho(b) = B. \tag{28}$$

Then by Section 3.2.1, B_3 action on $\mathcal{E}^\dagger$ is not $SL_2(\mathbb{Z})$ since $\operatorname{Inn}(F_2)$ action on enhanced holonomy is not trivial, if $j(\gamma) = \operatorname{Inn}(w)$, then $\gamma \cdot [(\operatorname{Inn}(T_{-(p)})(\rho), 0)] = \left[\left(\operatorname{Inn}(T_{-\rho(w)(p)})(\rho), 0\right)\right]$.

By Proposition 3.1, the action on logarithmic holonomy is given by $\Lambda : B_3 \longrightarrow SL_2(\mathbb{Z})$, with

$$\Lambda(\sigma_1) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} : \begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} \longmapsto \begin{pmatrix} \ell_1 \\ \ell_1 + \ell_2 \end{pmatrix}, \quad \Lambda(\sigma_2) = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} : \begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} \longmapsto \begin{pmatrix} \ell_1 - \ell_2 \\ \ell_2 \end{pmatrix}$$

and if we write $C := (\sigma_1\sigma_2\sigma_1)(\sigma_2\sigma_1\sigma_2) = (\sigma_1\sigma_2)^3$ and $Z := C^2$, then center is $Z(B_3) = \langle C \rangle$ and $\ker \Lambda = \langle Z \rangle = \langle C^2 \rangle$.

By Proposition 3.2, the subgroup $\langle Z \rangle$ acts freely on $\mathcal{E}^\dagger$. Therefore, by Proposition 3.3, the full B_3-action on $\mathcal{E}^\dagger$ is faithful.

By Lemma 3.4, the action $B_3 \curvearrowright \mathcal{T}^\dagger_{\mathrm{poly}}(S)$ is free.

3.1 Explicit Formula

For $[\rho,0]\in\mathcal{E}^\dagger$, $\rho(a)=(\lambda,U)$, $\rho(b)=(\mu,V)$, by Eqs. (24) and (26),

$$\sigma_1\cdot(\lambda,\mu,U,V)=(\lambda,\lambda\mu,U,U+\lambda V),\qquad \sigma_1^{-1}\cdot(\lambda,\mu,U,V)=\left(\lambda,\frac{\mu}{\lambda},U,\frac{V-U}{\lambda}\right), \tag{29}$$

$$\sigma_2\cdot(\lambda,\mu,U,V)=\left(\frac{\lambda}{\mu},\mu,\frac{U-V}{\mu},V\right),\qquad \sigma_2^{-1}\cdot(\lambda,\mu,U,V)=(\lambda\mu,\mu,\mu U+V,V). \tag{30}$$

For $q=[e_1,e_2;\ell_1,\ell_2]\in Y$,

$$\sigma_1\cdot q=[e_1+e^{\ell_1}e_2,e^{\ell_1}e_2;\ell_1,\ell_1+\ell_2],\qquad \sigma_1^{-1}\cdot q=[e_1-e_2,e^{-\ell_1}e_2;\ell_1,\ell_2-\ell_1], \tag{31}$$

$$\sigma_2\cdot q=[e^{-\ell_2}e_1,e_2-e_1;\ell_1-\ell_2,\ell_2],\qquad \sigma_2^{-1}\cdot q=[e^{\ell_2}e_1,e_2+e^{\ell_2}e_1;\ell_1+\ell_2,\ell_2]. \tag{32}$$

Directly compute give

$$j(Z)(a)=\partial_0 a\partial_0^{-1},\quad j(Z)(b)=\partial_0 b\partial_0^{-1}\implies j(Z)=\mathrm{Inn}_{F_2}(\partial_0). \tag{33}$$

Since $\rho(\partial_0)=T_\delta$, then for every $m\in\mathbb{Z}$,

$$\begin{aligned}Z^m\cdot[(\rho,p)]&=\left[\left(\rho\circ\mathrm{Inn}(\partial_0^{-m}),p\right)\right]=[(\rho(\partial_0)^{-m}\rho\,\rho(\partial_0)^m,p)]=\left[\left(T_\delta^{-m}\rho T_\delta^m,p\right)\right]\\&=[(\rho,T_\delta^m(p))]=[(\rho,p+m\delta)].\end{aligned} \tag{34}$$

Thus, on normalized enhanced holonomy coordinates,

$$Z^m\cdot(\lambda,\mu,U,V)=(\lambda,\mu,U+m(\lambda-1)\delta,V+m(\mu-1)\delta)\,,\quad \delta=\frac{(1-\mu)U+(\lambda-1)V}{\lambda\mu}. \tag{35}$$

Consequently, in room coordinates,

$$Z^m\cdot[e_1,e_2;\ell_1,\ell_2]=[e_1+m\lambda(\mu-1)\delta,e_2+m\mu(\lambda-1)\delta;\ell_1,\ell_2]\,,\quad \delta(q)=(1-e^{-\ell_1})e_1+(e^{-\ell_2}-1)e_2. \tag{36}$$

3.2 Property of Relative MCG action

3.2.1 Action on enhanced holonomy is not $SL_2(\mathbb{Z})$

Consider inner automorphism of F_2 , $\mathrm{Inn}(w)(x)=wxw^{-1}$, action on the ordinary character variety $\mathrm{Hom}\,(F_2,G)\,/G$, $\rho\sim g\rho g^{-1}$ is given by $\mathrm{Inn}(w)\cdot[\rho]=[\rho\circ\mathrm{Inn}(w)^{-1}]=[\rho]$ since $(\rho\circ\mathrm{Inn}(w)^{-1})\,(x)=\rho\,(w^{-1}xw)=\rho(w)^{-1}\rho(x)\rho(w)$. So the mapping-class action on the ordinary character variety depends only on $\mathrm{Out}\,(F_2)=\mathrm{Aut}\,(F_2)\,/\,\mathrm{Inn}\,(F_2)$ and $\mathrm{out}^+\,(F_2)\cong SL_2(\mathbb{Z})$.

But $\mathrm{Inn}\,(F_2)$ action on enhanced holonomy is not trivial, if $j(\gamma)=\mathrm{Inn}(w)$, then

$$\begin{aligned}\gamma\cdot[(\mathrm{Inn}(T_{-(p)})(\rho),0)]=\gamma\cdot[(\rho,p)]&=\left[\left(\rho\circ\mathrm{Inn}(w)^{-1},p\right)\right]\\&=[(\rho,\rho(w)(p))]\\&=\left[\left(T_{-\rho(w)(p)}\circ\rho\circ T_{\rho(w)(p)},0\right)\right]=\left[\left(\mathrm{Inn}(T_{-\rho(w)(p)})(\rho),0\right)\right].\end{aligned}$$

In normalized enhanced holonomy coordinates,

$$[(\rho,p)]=[\lambda,\mu;U,V;0],\quad U=A(p)-p,\quad V=B(p)-p.$$

and

$$[(\rho,\rho(w)(p))]=[\lambda,\mu;U_w,V_w;0],\quad U_w:=A(\rho(w)(p))-\rho(w)(p),\quad V_w:=B(\rho(w)(p))-\rho(w)(p).$$

we can also write

$$[\lambda,\mu;U_w,V_w;0]=[\lambda,\mu;U+(\lambda-1)\Delta_w,V+(\mu-1)\Delta_w;0]\,,\quad \Delta_w:=\rho(w)(p)-p.$$

Thus an inner automorphism generally changes the normalized enhanced holonomy coordinates, although it acts trivially on the ordinary character variety.

3.2.2 Action on logarithmic holonomy have kernel $\langle Z\rangle$ wich is a subgroup of center $\langle C\rangle$

For $[(\rho,p)]\in\mathcal{E}^\dagger$, write $\mathrm{Lin}(z\mapsto rz+c):=r$ define the logarithmic linear holonomy by

$$\ell(\rho):=\begin{pmatrix}\log\mathrm{Lin}(\rho(a))\\ \log\mathrm{Lin}(\rho(b))\end{pmatrix}=\begin{pmatrix}\ell_1\\ \ell_2\end{pmatrix}\in\mathbb{R}^2\setminus\{0\},$$

The action of B_3 on the logarithmic linear holonomy defines a representation

$$\Lambda: B_3\longrightarrow SL_2(\mathbb{Z})$$

by

$$\gamma\cdot\ell(\rho):=\ell\Big(\rho\circ j(\gamma)^{-1}\Big)=\Lambda(\gamma)\ell(\rho). \tag{37}$$

Indeed, for the two generators, one has

$$j(\sigma_1)^{-1}(a)=a,\qquad j(\sigma_1)^{-1}(b)=ab,\qquad j(\sigma_2)^{-1}(a)=b^{-1}a,\qquad j(\sigma_2)^{-1}(b)=b.$$

Hence

$$\begin{aligned}&(\rho\circ j(\sigma_1)^{-1})(a)=\rho(a), &&(\rho\circ j(\sigma_1)^{-1})(b)=\rho(a)\rho(b),\\ &(\rho\circ j(\sigma_2)^{-1})(a)=\rho(b)^{-1}\rho(a), &&(\rho\circ j(\sigma_2)^{-1})(b)=\rho(b).\end{aligned}$$

Therefore

$$\Lambda(\sigma_1)=\begin{pmatrix}1&0\\1&1\end{pmatrix}:\begin{pmatrix}\ell_1\\ \ell_2\end{pmatrix}\longmapsto\begin{pmatrix}\ell_1\\ \ell_1+\ell_2\end{pmatrix},\qquad \Lambda(\sigma_2)=\begin{pmatrix}1&-1\\0&1\end{pmatrix}:\begin{pmatrix}\ell_1\\ \ell_2\end{pmatrix}\longmapsto\begin{pmatrix}\ell_1-\ell_2\\ \ell_2\end{pmatrix}. \tag{38}$$

Proposition 3.1. *Define*

$$C:=(\sigma_1\sigma_2)^3=(\sigma_2\sigma_1)^3,\qquad Z:=C^2. \tag{39}$$

Then centre of B_3 is

$$Z(B_3)=\langle C\rangle,$$

and the kernel of the action on logarithmic holonomy is

$$\ker\Lambda=\langle Z\rangle=\langle C^2\rangle\subsetneq Z(B_3).$$

Proof. We are going to show

$$\ker\Lambda=\{\gamma\in B_3\mid\Lambda(\gamma)=I\}=\langle Z\rangle=\{Z^m\mid m\in\mathbb{Z}\},$$

First, since $B_3=\langle\sigma_1,\sigma_2\mid\sigma_1\sigma_2\sigma_1=\sigma_2\sigma_1\sigma_2\rangle$, we directly verify $\Lambda(\sigma_1\sigma_2\sigma_1)=\Lambda(\sigma_2\sigma_1\sigma_2)=\begin{pmatrix}0&-1\\1&0\end{pmatrix}$. Denote $C:=(\sigma_1\sigma_2\sigma_1)(\sigma_2\sigma_1\sigma_2)=(\sigma_1\sigma_2\sigma_1)^2=(\sigma_1\sigma_2)^3$. Then we have

$$\Lambda(C)=-I\implies\Lambda(Z)=I\implies\langle Z\rangle\subseteq\ker\Lambda.$$

Moreover,

$$x:=\sigma_1\sigma_2\sigma_1,\qquad y:=\sigma_1\sigma_2,\qquad \sigma_1=y^{-1}x,\qquad \sigma_2=x^{-1}y^2.$$

then the braid group could write as

$$B_3\cong\Big\langle x,y\;\Big|\;x^2=y^3\Big\rangle,\qquad x^2=y^3=C.$$

For the reverse inclusion, compose Λ with $SL_2(\mathbb{Z})\longrightarrow PSL_2(\mathbb{Z})$ gives homomorphism

$$\overline{\Lambda}: B_3\longrightarrow PSL_2(\mathbb{Z}).$$

Therefore

$$B_3/\langle C\rangle \cong \left\langle x, y \;\middle|\; x^2 = 1,\; y^3 = 1 \right\rangle \cong PSL_2(\mathbb{Z}) \implies \ker \overline{\Lambda} = \langle C\rangle.$$

Otherwise, since the center of $PSL_2(\mathbb{Z})$ is trivial, we have $Z(B_3) \subseteq \langle C\rangle$. Since $C = x^2 = y^3$, it commutes with both x and y, hence $\langle C\rangle \subseteq Z(B_3)$. Therefore $Z(B_3) = \langle C\rangle$.

We are going to show

$$\Lambda(\gamma) = I \quad\implies\quad \gamma = C^k \quad\implies\quad k \text{ even} \quad\implies\quad \gamma = Z^m.$$

Now let $\gamma \in \ker \Lambda$. Then $\gamma \in \ker \overline{\Lambda} = \langle C\rangle$, so $\gamma = C^k$ for some $k \in \mathbb{Z}$. Since

$$I = \Lambda(\gamma) = \Lambda(C)^k = (-I)^k,$$

k is even. Writing $k = 2m$, we obtain $\gamma = C^{2m} = Z^m$. $\square$

3.2.3 The center subgroup $\langle Z\rangle$ acts freely on enhanced holonomy

We shall prove that the central subgroup $\langle Z\rangle$ acts freely. This does not imply that the full B_3-action is free, but together with $\ker \Lambda = \langle Z\rangle$, it will imply that the B_3-action on $\mathcal{E}^\dagger$ is faithful.

Proposition 3.2 (The central subgroup acts freely)**.** *The central subgroup*

$$\langle Z\rangle \curvearrowright \mathcal{E}^\dagger$$

acts freely.

Proof. By 34, suppose

$$Z^m \cdot [(\rho, p)] = [(\rho, p + m\delta)] = [(\rho, p)].$$

Then there exists $g(z) = rz + c$ such that

$$g\rho g^{-1} = \rho, \qquad g(p) = p + m\delta.$$

Since g commutes with $\rho(\partial_0) = T_\delta$, so we have

$$T_\delta = gT_\delta g^{-1} = T_{r\delta}.$$

As $\delta \neq 0$, we get $r = 1$, so $g = T_c$. Write $A(z) = \lambda z + u$, $B(z) = \mu z + v$. Then relations $gA = Ag$ and $gB = Bg$ give

$$(\lambda - 1)c = 0, \qquad (\mu - 1)c = 0.$$

Since $\ell(\rho) \neq 0$, at least one of λ, μ is different from 1, hence $c = 0$. Therefore $g = \mathrm{id}$, so

$$p = p + m\delta.$$

Since $\delta \neq 0$, we obtain $m = 0$. $\square$

3.2.4 Action on enhanced holonomy is faithful

Proposition 3.3 (Faithfulness of the B_3-action on enhanced holonomy)**.** *The B_3-action on $\mathcal{E}^\dagger$ is faithful.*

Proof. Suppose that $\gamma \in B_3$ acts trivially on every point of $\mathcal{E}^\dagger$. It then fixes every logarithmic holonomy vector, so $\Lambda(\gamma) = I$.

By Proposition 3.1, there exists $m \in \mathbb{Z}$ such that $\gamma = Z^m$.

By Proposition 3.2, the subgroup $\langle Z\rangle$ acts freely on $\mathcal{E}^\dagger$, so $m = 0$, thus $\gamma = 1$. $\square$

3.2.5 Action on Teichmüller space is free

Lemma 3.4 (Freeness of the change-of-marking action)**.** *The action*

$$B_3 \curvearrowright \mathcal{T}^{\dagger}_{\mathrm{poly}}(S)$$

is free.

Proof. Suppose that

$$\gamma \cdot [X, f] = [X, f \circ \gamma^{-1}] = [X, f].$$

then there exists a dilation automorphism $F : X \to X$ such that

$$F \circ f \circ \gamma^{-1} \simeq f \iff f^{-1} \circ F \circ f \simeq \gamma.$$

Take two copies of X and glue them along their boundary. This gives a closed Riemann surface $\widehat{X} = X \cup_{\partial X} \overline{X}$, and the boundary preserving conformal automorphism F extends to a conformal automorphism

$$\widehat{F}(x) = \begin{cases} F(x), & x \in X, \\ \overline{F}(x), & x \in \overline{X}, \end{cases}$$

Since the conformal automorphism group of a closed Riemann surface of genus g at least two is at most $84(g-1)$ by [3, Theorem 7.4]. Therefore there exists $N > 0$ such that $\widehat{F}^N = \mathrm{id}_{\widehat{X}}$. Restricting to X, we obtain

$$F^N = \mathrm{id}_X .$$

we then have

$$\gamma^N \simeq (f^{-1} \circ F \circ f)^N = f^{-1} \circ F^N \circ f = \mathrm{id}_S .$$

Hence $\gamma^N = 1$ in B_3. Since the relative mapping class group $B_3 \cong \mathrm{Mod}(S, \partial S)$ is torsion-free by [3, Corollary 7.3] , we have $\gamma = 1$.

□

4 Labelled cut systems and explicit room changes

4.0.1 Labelled cut systems and Corner words

Definition 4.1 (Oriented cut loop)**.** An **oriented cut loop** based at s is an oriented simple loop in S based at s, whose interior is contained in $\mathrm{int}(S)$.

Definition 4.2 (Labelled cut pair)**.** A **labelled cut pair** is an ordered pair (α, β) of oriented cut loops with disjoint interiors such that cutting S along $\alpha \cup \beta$ produces a topological disk $D_{\alpha,\beta}$ whose directed boundary sides, in cyclic order, are

$$\alpha^-, \quad \beta^+, \quad (\alpha^+)^{-1}, \quad (\beta^-)^{-1}, \quad \varepsilon,$$

where $\alpha^\pm$ and $\beta^\pm$ are the two copies of the cut loops and ε descends to ∂S.

Fix a lift $\widetilde{s}_0$ of s. Let $\kappa = (\alpha, \beta)$ be a labelled cut pair, and lift $D_{\alpha,\beta}$ so that the initial corner of the directed side α^- is $\widetilde{s}_0$. Following the directed boundary of $D_{\alpha,\beta}$ in the order $(\alpha^-, \beta^+, (\alpha^+)^{-1}, (\beta^-)^{-1}, \varepsilon)$, we obtain five ordered corner lifts

$$w_0^\kappa \widetilde{s}_0, \ldots, w_4^\kappa \widetilde{s}_0, \qquad w_0^\kappa = 1.$$

Definition 4.3 (Corner words)**.** The ordered tuple

$$\mathbf{w}^\kappa = (w_0^\kappa, \ldots, w_4^\kappa)$$

is called the **corner words** of κ, relative to the fixed lift $\widetilde{s}_0$.

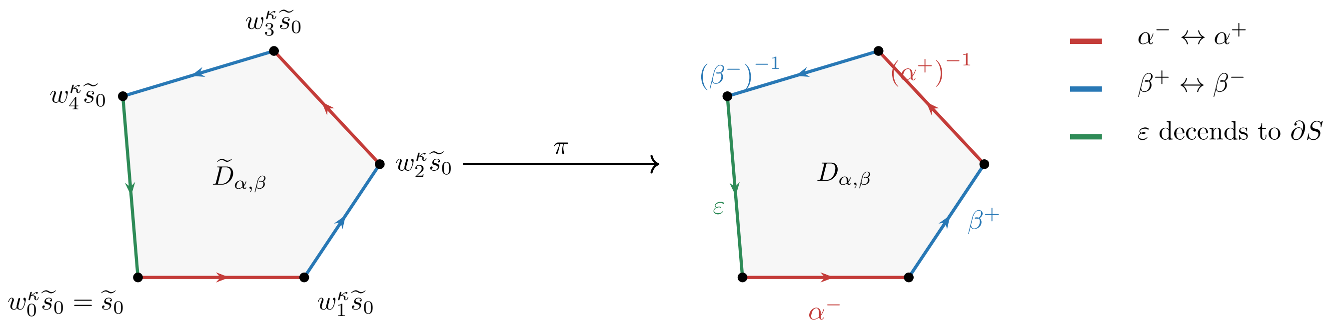


Figure 1: A lifted cut disk and its projected cut disk. The ordered corner lifts $w_0^\kappa \widetilde{s}_0, \ldots, w_4^\kappa \widetilde{s}_0$ are read by following the directed boundary of $D_{\alpha,\beta}$ in the order $\alpha^-, \beta^+, (\alpha^+)^{-1}, (\beta^-)^{-1}, \varepsilon$.

Definition 4.4 (Labelled cut system)**.** Two labelled cut pairs (α, β) and (α', β') are *equivalent* if they are jointly ambient isotopic relative to ∂S, through labelled oriented cut pairs.

Explicitly, two labelled cut pairs (α, β) and (α', β') are equivalent if there exists an isotopy

$$H_t : S \longrightarrow S, \qquad 0 \le t \le 1,$$

through homeomorphisms such that

$$H_0 = \mathrm{id}_S, \qquad H_t|_{\partial S} = \mathrm{id}_{\partial S}, \qquad H_1(\alpha, \beta) = (\alpha', \beta'),$$

with the labels and orientations preserved throughout the isotopy.

A *labelled cut system* is an equivalence class $\kappa = [\alpha, \beta]_{\mathrm{cut}}$. For a chosen representative, we write $\kappa = (\alpha_\kappa, \beta_\kappa)$.

Let $\mathcal{K}(S)$ denote the set of labelled cut systems.

Definition 4.5 (Strict realization)**.** A labelled cut system κ is *strictly realizable* on X if it admits a lifted cut disk $\widetilde{D}_\kappa$ such that $\widehat{\mathrm{dev}}(\widetilde{D}_\kappa)$ is a strictly convex labelled pentagon.

4.0.2 MCG action on cut systems is transitive

Lemma 4.6 (Mapping class group acts transitively)**.** *The relative mapping class group acts transitively on $\mathcal{K}(S)$. Hence, after fixing a standard cut system κ_0, every labelled cut system can be written as $\kappa = \gamma\kappa_0$ for some $\gamma \in B_3$.*

Proof. Let $\kappa, \kappa' \in \mathcal{K}(S)$. Cutting S along either system produces a labelled pentagonal disk with the same cyclic side pattern.

A homeomorphism between the two disks preserving the labels, directions, and side pairings descends after regluing to an orientation-preserving homeomorphism of S, fixed on ∂S, sending κ to κ'. □

Let $h_\gamma : S \to S$ represent $\gamma \in \Gamma_\partial$, and let $\widetilde{h}_\gamma : \widetilde{S} \to \widetilde{S}$ be the lift fixing $\widetilde{s}_0$. Then

$$\widetilde{h}_\gamma(g\widetilde{x}) = j(\gamma)(g)\widetilde{h}_\gamma(\widetilde{x}).$$

Then for every $\gamma \in \Gamma_\partial$, we have

$$w_i^{\gamma\kappa} = j(\gamma)(w_i^{\kappa}), \qquad 0 \le i \le 4,$$

Let $u_\kappa, v_\kappa \in F_2$ denote the side deck transformations defined by

$$u_\kappa(\alpha_\kappa^-) = \alpha_\kappa^+, \qquad v_\kappa(\beta_\kappa^-) = \beta_\kappa^+.$$

then we have

$$u_{\gamma\kappa} = j(\gamma)(u_\kappa), \qquad v_{\gamma\kappa} = j(\gamma)(v_\kappa).$$

Example 4.1 (The standard pentagonal cut system). For the standard pentagonal cut system κ_0, the corner words and labelled side deck transformations are

$$\mathbf{w} = (w_0, w_1, w_2, w_3, w_4) = (1, a^{-1}ba, ba, a, \partial_0^{-1}), \qquad (u_{\kappa_0}, v_{\kappa_0}) = (a, b).$$

Then we have

$$j(\sigma_1^{-1})(\mathbf{w}) = (1, ba, aba, a, \partial_0^{-1}), \qquad (u_{\sigma_1^{-1}\kappa_0}, v_{\sigma_1^{-1}\kappa_0}) = (a, ab).$$

$$j(\sigma_2^{-1})(\mathbf{w}) = (1, a^{-1}ba, a, b^{-1}a, \partial_0^{-1}), \qquad (u_{\sigma_2^{-1}\kappa_0}, v_{\sigma_2^{-1}\kappa_0}) = (b^{-1}a, b).$$

$$j(Z^{-1})(\mathbf{w}) = (1,\, \partial_0^{-1}a^{-1}ba\partial_0,\, \partial_0^{-1}ba\partial_0,\, \partial_0^{-1}a\partial_0,\, \partial_0^{-1}), \qquad (u_{Z^{-1}\kappa_0}, v_{Z^{-1}\kappa_0}) = (\partial_0^{-1}a\partial_0,\, \partial_0^{-1}b\partial_0).$$

Remark 4.7. Aut (F_2) transitive action on basis space, like for another basis (u, v) we only need automorphism : $a \mapsto u, b \mapsto v$.

But $\Gamma_\partial \cong \{\phi \in \operatorname{Aut}(F_2) \mid \phi(\partial_0) = \partial_0\}$, which means when $\partial_0 = ba^{-1}[a,b](ba)$, for $\gamma \in \Gamma_\partial$, such that $u = j(\gamma)(a), v = j(\gamma)(b)$, we also need ask

$$(vu)^{-1}[u,v](vu) = j(\gamma)((ba)^{-1}[a,b](ba)) = j(\gamma)(\partial_0) = \partial_0 = (ba)^{-1}[a,b](ba).$$

Accordingly, the basis of F_2 arising from labelled cut systems lie in

$$\mathscr{B}_{\partial_0} = \left\{(u,v) \in F_2^2 \mid F_2 = \langle u, v\rangle, (vu)^{-1}[u,v](vu) = \partial_0\right\}.$$

4.0.3 Room charts and change of cut system

Let $q \in \mathcal{Y}_{++}$ and let κ be a labelled cut system. Let $P(q)$ be the labelled pentagonal room associated to q, and let $\pi_q : P(q) \to S(q)$ be the quotient map obtained by gluing the paired sides.

Let $\pi_\kappa : D_\kappa \to S$ be the quotient map of the cut-open disk. Choose a label-, direction-, and pairing-preserving homeomorphism $\varphi_{\kappa,q} : D_\kappa \to P(q)$. Since $\varphi_{\kappa,q}$ respects the side-pairing equivalence relations, it induces a unique orientation-preserving homeomorphism $f_{\kappa,q} : S \to S(q)$ satisfying

$$f_{\kappa,q} \circ \pi_\kappa = \pi_q \circ \varphi_{\kappa,q}.$$

Equivalently, the following diagram commutes:

$$\begin{tikzcd}
D_\kappa \arrow[r, "\varphi_{\kappa,q}"] \arrow[d, "\pi_\kappa"'] & P(q) \arrow[d, "\pi_q"] \\
S \arrow[r, "f_{\kappa,q}"'] & S(q).
\end{tikzcd}$$

Definition 4.8 (Room chart associated to a labelled cut system)**.** We call $f_{\kappa,q}$ the marking induced by κ, and define

$$\Theta_\kappa : \mathcal{Y}_{++} \longrightarrow \mathcal{T}^{\dagger}_{\text{poly}}(S), \qquad \Theta_\kappa(q) := [S(q), f_{\kappa,q}].$$

Denote $\mathcal{U}_\kappa$ as the locus of marked dilation surfaces on which κ is strictly realizable, it is clear that

$$\operatorname{Im}(\Theta_\kappa) \subset \mathcal{U}_\kappa.$$

We are going to show that

$$\Theta_\kappa : \mathcal{Y}_{++} \longrightarrow \mathcal{U}_\kappa$$

is a bijection:

Lemma 4.9 (Five-corner recovery)**.** *For every labelled cut system κ, the map*

$$\Theta_\kappa : \mathcal{Y}_{++} \longrightarrow \mathcal{U}_\kappa$$

is a diffeomorphism.

Proof. **Surjectivity.** Assume $X \in \mathcal{U}_\kappa$ with development $(\widehat{\text{dev}}, \rho)$, we wish to find the room realize κ. Denote the developed vertices are

$$P_i^\kappa = \widehat{\text{dev}}(w_i^\kappa \tilde{s}_0) = \rho(w_i^\kappa)(0).$$

and two labelled side-pairing maps are

$$A_\kappa := \rho(u_\kappa) : z \to \lambda_\kappa z + U_\kappa, \qquad B_\kappa := \rho(v_\kappa) : z \to \mu_\kappa z + V_\kappa.$$

Thus the room determined by (X, κ) has normalized enhanced-holonomy coordinates

$$\eta_\kappa(X) = (\lambda_\kappa, \mu_\kappa, U_\kappa, V_\kappa).$$

Denote $\ell_1^\kappa = \log \lambda_\kappa, \ell_2^\kappa = \log \mu_\kappa$, Its room coordinates are

$$q_\kappa(X) = [e_1^\kappa, e_2^\kappa; \ell_1^\kappa, \ell_2^\kappa],$$

where

$$e_1^\kappa = P_2^\kappa - P_3^\kappa = V_\kappa + (\mu_\kappa - 1)U_\kappa, \qquad e_2^\kappa = P_2^\kappa - P_1^\kappa = U_\kappa + (1 - \lambda_\kappa^{-1})e_1^\kappa$$

Injectivity. Suppose

$$\Theta_\kappa(q) = \Theta_\kappa(q') = X.$$

For the fixed cut system κ, the five developed corners of X have same developed vertex

$$P_i^\kappa = \widehat{\mathrm{dev}}(w_i^\kappa \widetilde{s}_0).$$

Hence both q and q' both satisfy

$$e_1 = P_2^\kappa - P_3^\kappa, \qquad e_2 = P_2^\kappa - P_1^\kappa,$$

and

$$P_1^\kappa = \lambda^{-1} e_1, \qquad P_3^\kappa - P_4^\kappa = \mu^{-1} e_2.$$

Thus the five labelled corners uniquely determine e_1, e_2, λ, μ, and hence

$$q = q' = [e_1, e_2; \ell_1, \ell_2] \in \mathcal{Y}_{++}.$$

$\square$

Since $\Theta_\kappa : \mathcal{Y}_{++} \to \mathcal{U}_\kappa$ is a bijection, for $X \in \mathcal{U}_\kappa$ define

$$q_\kappa(X) := \Theta_\kappa^{-1}(X), \qquad \eta_\kappa(X) := \Phi(q_\kappa(X)).$$

We now show that, for a fixed marked surface, the action on room coordinates is exactly induced by changing the cut system.

Proposition 4.10 (Change of cut system)**.** *Let X be a marked dilation surface, and suppose that κ_0 and $\gamma^{-1}\kappa_0$ are both strictly realizable on X. Then*

$$q_{\gamma^{-1}\kappa_0}(X) = \gamma \cdot q_{\kappa_0}(X).$$

Proof. Write

$$\eta_{\kappa_0}(X) = [(\rho, 0)].$$

When the cut system changes from κ_0 to $\gamma^{-1}\kappa_0$, its corner words and side-pairing words change by $j(\gamma)^{-1}$. Hence

$$\eta_{\gamma^{-1}\kappa_0}(X) = \Big[\Big(\rho \circ j(\gamma)^{-1}, 0\Big)\Big] = \gamma \cdot \eta_{\kappa_0}(X).$$

Therefore

$$q_{\gamma^{-1}\kappa_0}(X) = \Phi^{-1}(\eta_{\gamma^{-1}\kappa_0}(X)) = \Phi^{-1}(\gamma \cdot \eta_{\kappa_0}(X)) = \gamma \cdot q_{\kappa_0}(X).$$

$\square$

For a fixed room parameter $q \in \mathcal{Y}_{++}$, all points

$$\Theta_\kappa(q) = [S(q), f_{\kappa,q}]$$

have the same underlying dilation surface $S(q)$. The change of marking is exactly induced by the corresponding change of cut system:

Lemma 4.11 (Fixed room and different cut systems)**.** *Let $q \in \mathcal{Y}_{++}$, let κ be a labelled cut system, and let $\gamma \in \Gamma_\partial$. Then*

$$\Theta_{\gamma^{-1}\kappa}(q) = \gamma^{-1} \cdot \Theta_\kappa(q).$$

In particular,

$$\Theta_{\gamma^{-1}\kappa}(q) = \Theta_\kappa(q) \Longleftrightarrow \gamma = 1.$$

Proof. Let $h_\gamma : S \longrightarrow S$ be a homeomorphism representing $\gamma \in \Gamma_\partial$, and let

$$h_\gamma^D : D_{\gamma^{-1}\kappa} \longrightarrow D_\kappa$$

be the induced homeomorphism between the cut-open disks. Since $h_\gamma(\gamma^{-1}\kappa) = \kappa$, the map h_γ^D preserves the labels, orientations, and side pairings. Let

$$\varphi_{\kappa,q} : D_\kappa \longrightarrow P(q)$$

be the polygon identification which descends to the marking $f_{\kappa,q}$. Define

$$\varphi_{\gamma^{-1}\kappa,q} := \varphi_{\kappa,q} \circ h_\gamma^D : D_{\gamma^{-1}\kappa} \longrightarrow P(q).$$

Passing to the side-pairing quotients gives

$$f_{\gamma^{-1}\kappa,q} = f_{\kappa,q} \circ h_\gamma.$$

Therefore

$$\Theta_{\gamma^{-1}\kappa}(q) = [S(q), f_{\gamma^{-1}\kappa,q}] = [S(q), f_{\kappa,q} \circ h_\gamma] = \gamma^{-1} \cdot [S(q), f_{\kappa,q}] = \gamma^{-1} \cdot \Theta_\kappa(q).$$

Then if $\Theta_{\gamma^{-1}\kappa}(q) = \Theta_\kappa(q)$, we have γ^{-1} fixes the point $\Theta_\kappa(q) \in \mathcal{T}^\dagger_{\mathrm{poly}}(S)$. By Lemma 3.4, the B_3-action on $\mathcal{T}^\dagger_{\mathrm{poly}}(S)$ is free, so $\gamma = 1$. □

Fix X and a developing pair. If both κ_0 and $\gamma^{-1}\kappa_0$ are strictly realizable on X, then by Proposition 4.10,

$$q_{\gamma^{-1}\kappa_0}(X) = \gamma \cdot q_{\kappa_0}(X).$$

Conversely, for $X = \Theta_{\kappa_0}(q)$, when $\gamma q \in \mathcal{Y}_{++}$, is it arise from the cut system $\gamma^{-1}\kappa_0$ on the same surface X ? Equivalently, do we have

$$\Theta_{\kappa_0}(q) \stackrel{?}{=} \Theta_{\gamma^{-1}\kappa_0}(\gamma q).$$

The central example in Section 6 considers $(q, \gamma) = (q_0, Z)$, and found $\Theta_{\kappa_0}(q_0) \neq \Theta_{Z^{-1}\kappa_0}(Z q_0)$.

5 Room charts for the Teichmüller space

Lemma 5.1 (Holonomy in room charts)**.** *Let κ_0 be the standard labelled cut system.*

1. *For every $q \in \mathcal{Y}_{++}$,*

$$\widehat{\mathrm{Hol}}(\Theta_{\kappa_0}(q)) = \Phi(q).$$

   Equivalently, we have $\widehat{\mathrm{Hol}} \circ \Theta_{\kappa_0} = \Phi$.

2. *If $\kappa = \gamma\kappa_0$ is a labelled cut system, then*

$$\widehat{\mathrm{Hol}}\big|_{\mathcal{U}_\kappa} : \mathcal{U}_\kappa \longrightarrow \gamma \cdot \mathcal{E}_{++}$$

   is a diffeomorphism.

Proof. **(1)** Let

$$\Theta_{\kappa_0}(q) = [S(q), f_{\kappa_0,q}].$$

By construction, $S(q)$ is obtained from the room $P(q)$ using the side-pairing maps

$$\rho(a) = A, \qquad \rho(b) = B.$$

Moreover, the distinguished lift $\widetilde{s}_0$ develops to $P_0 = 0$. Hence

$$\widehat{\mathrm{Hol}}(\Theta_{\kappa_0}(q)) = [(\rho, 0)] = \Phi(q).$$

(2) By part (1), $\widehat{\mathrm{Hol}} \circ \Theta_{\kappa_0} = \Phi$. Since Θ_{κ_0} and Φ are diffeomorphisms,

$$\widehat{\mathrm{Hol}}|_{\mathcal{U}_{\kappa_0}} = \Phi \circ \Theta_{\kappa_0}^{-1}$$

is a diffeomorphism onto $\mathcal{E}_{++}$. If $\kappa = \gamma\kappa_0$, then by Lemma 4.11, $\Theta_{\gamma\kappa_0} = \gamma \cdot \Theta_{\kappa_0}$, taking images gives

$$\mathcal{U}_{\gamma\kappa_0} = \gamma \cdot \mathcal{U}_{\kappa_0}.$$

and by equivariance,

$$\widehat{\mathrm{Hol}}(\gamma \cdot X) = \gamma \cdot \widehat{\mathrm{Hol}}(X).$$

Therefore

$$\widehat{\mathrm{Hol}}(\mathcal{U}_\kappa) = \gamma \cdot \mathcal{E}_{++},$$

□

Lemma 5.2 (Strictification of a weak room)**.** *Let $q \in \mathcal{Y}_+$ have logarithmic multipliers*

$$(t, 0) \qquad \textit{or} \qquad (0, t), \qquad t > 0.$$

Then the same marked dilation surface admits a strictly convex room, for another labelled cut system, with logarithmic multipliers

$$(t, t).$$

Proof. If q has logarithmic multipliers $(t, 0)$ or $(0, t)$, apply the σ_1-step or σ_2^{-1}-step. By Eq. (31), the new room has logarithmic multipliers (t, t). Hence it is strictly convex by Eq. (5).

In both cases, the step is a cut-and-paste operation of convex room, so the marked dilation surface is unchanged and only the labelled cut system changes. □

Proposition 5.3 (Room-chart cover)**.** *The strict room charts cover the polygonable deformation space:*

$$\mathcal{T}^{\dagger}_{\mathrm{poly}}(S) = \bigcup_{\kappa \in \mathcal{K}(S)} \mathcal{U}_\kappa.$$

Consequently,

$$\widehat{\mathrm{Hol}} : \mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \longrightarrow \mathcal{E}^{\dagger}$$

is a local diffeomorphism.

Proof. Let

$$X = [M, f] \in \mathcal{T}^{\dagger}_{\mathrm{poly}}(S).$$

then M admits an embedded convex pentagonal room by [1, Lemma 3.1]. Then its logarithmic multipliers satisfy $(\ell_1, \ell_2) \in \mathbb{R}^2_{\geq 0} \setminus \{(0, 0)\}$.

If $\ell_1, \ell_2 > 0$, the convex room on M is strictly convex by Eq. (5). Let $\overline{\kappa}$ be the labelled cut system on M determined by this room. Pull it back by the marking and suppose $f(\kappa) = \overline{\kappa}$, then $X \in \mathcal{U}_\kappa$.

If exactly one of ℓ_1, ℓ_2 is zero, the room on M is weakly convex by Eq. (6). By Lemma 5.2, the same dilation surface M admits a strictly convex room for another labelled cut system $\overline{\kappa}'$ on M. Pulling this cut system back by the marking and suppose $\kappa' := f^{-1}(\overline{\kappa}')$. Then $X \in \mathcal{U}_{\kappa'}$.

Therefore $\mathcal{T}^{\dagger}_{\mathrm{poly}}(S) = \bigcup_{\kappa \in \mathcal{K}(S)} \mathcal{U}_\kappa$.

Now let $X \in \mathcal{U}_\kappa$, and write $\kappa = \gamma\kappa_0$. By Lemma 5.1, we have $\widehat{\mathrm{Hol}}|_{\mathcal{U}_\kappa} : \mathcal{U}_\kappa \longrightarrow \gamma \cdot \mathcal{E}_{++}$ is a diffeomorphism. Hence $\widehat{\mathrm{Hol}}$ is a local diffeomorphism. □

6 The motivating central example

6.1 The geometric meaning of the central comparison

When we have a $q_0 \in \mathcal{Y}_{++}$ we can draw a pentagon, and we can glue to a surface.

$$q \xmapsto{\Theta_{\kappa_0}} X_0.$$

Since $\widehat{\mathrm{Hol}} \circ \Theta_{\kappa_0} = \Phi|_{\mathcal{Y}_{++}}$ we know the surface have the same enhanced holonomy $\eta := \Phi(q_0)$.

But when we have this old surface X_0 , and then for some formal room Zq_0 with the different enhanced holonomy $Z\eta$could we say it still a room for X_0 through some other cut system $\gamma\kappa_0$?

The real question turned to be

Could Zq_0 realize a cut system forX_0 through the cut system $Z^{-1}\kappa_0$?

Equivalently, one asks whether

$$\Theta_{Z^{-1}\kappa_0}(Zq_0) = \Theta_{\kappa_0}(q_0) \Longleftrightarrow X_1 = Z^{-1}\Theta_{\kappa_0}(Zq_0) = \Theta_{\kappa_0}(q_0) = X_0.$$

If the answer is yes, then Zq_0 must be the developed image of an actual lifted cut disk on X_0 representing the cut-system class $Z^{-1}\kappa_0$, If $w_0, \ldots, w_4$ are the corner words of the standard cut system κ_0, then the corner words of $Z^{-1}\kappa_0$ are

$$L_i = j(Z^{-1})(w_i) = \partial_0^{-1} w_i \partial_0.$$

Hence the lifted cut disk must have corners $L_0\widetilde{s}_0, \ldots, L_4\widetilde{s}_0$, in this prescribed cyclic order. Denote five vertices of Zq_0 with $Q_0, \ldots, Q_4$ then we have

$$\widehat{\mathrm{dev}}\,(L_i\tilde{s}_0) = Q_i,$$

And every consecutive pair $L_i\widetilde{s}_0, L_{i+1}\widetilde{s}_0$ must be joined by a arc $[L_i\widetilde{s}_0, L_{i+1}\widetilde{s}_0]$ contain no singularities such that

$$\widehat{\mathrm{dev}}\,([L_i\widetilde{s}_0, L_{i+1}\widetilde{s}_0]) = [Q_i, Q_{i+1}],$$

6.2 Simplest Example

Consider the simplest convex room example with basis as a square and dilation parameter are 2:

$$q_0 = [1, i; \log 2, \log 2] \in \mathcal{Y}_{++}, \qquad X_0 := \Theta_{\kappa_0}(q_0), \qquad \eta_0 := \Phi(q_0). \tag{40}$$

Its vertices and side pairings are

$$P_0 = 0, \quad P_1 = \frac{1}{2}, \quad P_2 = \frac{1}{2} + i, \quad P_3 = -\frac{1}{2} + i, \quad P_4 = -\frac{1}{2} + \frac{i}{2}, \tag{41}$$

$$A(z) = 2z - \frac{1}{2} + i, \qquad B(z) = 2z + \frac{3}{2} - i. \tag{42}$$

$$\rho(\partial_0)(z) = z + \delta, \qquad \delta = \frac{1}{2} - \frac{i}{2}, \qquad \partial_0 = a^{-1}b^{-1}ab. \tag{43}$$

For every $m \in \mathbb{Z}$, let **compensated branches** which defined before be

$$X_m := X_m(q_0) = Z^{-m}\Theta_{\kappa_0}(Z^m q_0).$$

This leads to two natural questions:

1. Does the fibre consist precisely of the central family?

$$\widehat{\mathrm{Hol}}^{-1}(\eta_0) = \{X_m \mid m \in \mathbb{Z}\}.$$

2. Are the members of this family pairwise distinct? That is,

$$X_m \neq X_n \qquad \text{whenever } m \neq n.$$

As a first step, we prove explicitly that

$$X_1 = Z^{-1}\Theta_{\kappa_0}(Zq_0) \neq \Theta_{\kappa_0}(q_0)X_0.$$

For $X_1 = Z^{-1}\Theta_{\kappa_0}(Zq_0)$, the vertices of the room $Zq_0 = [2-i, 1; \log 2, \log 2]$ are

$$Q_0 = 0, \quad Q_1 = 1 - \frac{i}{2}, \quad Q_2 = 2 - \frac{i}{2}, \quad Q_3 = \frac{i}{2}, \quad Q_4 = -\frac{1}{2} + \frac{i}{2}. \tag{44}$$

Fix lift singularity $\widetilde{s}_0$, the standard **corner words** with $\widehat{\operatorname{dev}}(w_i\widetilde{s}_0) = P_i$ are given by

$$w_0 = 1, \quad w_1 = a^{-1}ba, \quad w_2 = ba, \quad w_3 = a, \quad w_4 = \partial_0^{-1}, \tag{45}$$

the labels of $Z^{-1}\kappa_0$, action by $j(Z^{-1}) = \operatorname{Inn}(\partial_0^{-1})$, give $L_i = \partial_0^{-1}w_i\partial_0$ as new corner words

$$L_0 = 1, \quad L_1 = \partial_0^{-1}b, \quad L_2 = \partial_0^{-1}ab, \quad L_3 = \partial_0^{-1}b^{-1}ab, \quad L_4 = \partial_0^{-1}. \tag{46}$$

They develop to the Q_i, because

$$\widehat{\operatorname{dev}}(L_i\widetilde{s}_0) = \rho(\partial_0^{-1}w_i\partial_0)(0) = \rho(w_i)(\delta) - \delta = Q_i. \tag{47}$$

we then need there is a arc contain no lift of singularity and connect

$$d^{-1}b\widetilde{s}_0 \longrightarrow d^{-1}ab\widetilde{s}_0 \tag{48}$$

then developing to

$$\left[1 - \frac{i}{2}, 2 - \frac{i}{2}\right]. \tag{49}$$

Lemma 6.1 (Wrong-sheet obstruction). *The arc by Eqs. (48) and (49) does not exist on X_0. So*

$$X_1 = Z^{-1}\Theta_{\kappa_0}(Zq_0) \neq \Theta_{\kappa_0}(q_0)X_0.$$

Proof. The pentagon P is the original Euclidean room for X_0, determinedby q_0. Let $\widetilde{P}$ be the chosen lift so that its P_j-corner is $w_j\widetilde{s}_0$, which develops to P_j.

Suppose the room Zq_0 have Euclidean pentagon we denote by Q, could realize the cut system $Z^{-1}\kappa_0$ on X_0.

Then there would exist lift $\widetilde{Q}$ whose Q_j-corner is $\partial_0^{-1}w_j\partial_0\widetilde{s}_0$, developing to Q_j. and one side of the disk $\widetilde{Q}$ would have to be an arc joining $\partial_0^{-1}b\widetilde{s}_0$ to $\partial_0^{-1}ab\widetilde{s}_0$, with no singular lift in its interior, and developing to

$$[Q_1, Q_2] = \left[1 - \frac{i}{2}, 2 - \frac{i}{2}\right].$$

Thus, setting $x := \partial_0^{-1}b\widetilde{s}_0$, the required arc must leave x in the positive-real developed direction.

It's remarkable that $\widetilde{Q}$ and $\widetilde{P}$ all in the dilation covering space of X_0, but they need not differ by a deck transformation, and deck translates of $\widetilde{P}$ might only overlap $\widetilde{Q}$. Indeed, deck translates of $\widetilde{P}$ project to the same cut disk on X_0, whereas $\widetilde{Q}$ would represent $Z^{-1}\kappa_0$ so then another cut disk.

To determine whether such an arc exists, we examine a complete singular neighborhood U of singularity lift x in the lifted dilation structure of X_0.

Since the five corners of P are identified with the unique singularity of X_0, their lifts around the singular lift x belong to exactly five deck translates of $\widetilde{P}$, one for each corner type P_j. Indeed, if the P_j-corner of $\widetilde{P}$ is $w_j\widetilde{s}_0$, then the copy whose P_j-corner is $x = \partial_0^{-1}b\widetilde{s}_0$ is $g_j\widetilde{P}, g_j := \partial_0^{-1}bw_j^{-1}$, since $g_jw_j\widetilde{s}_0 = x$. Under the developing map, each $g_j\widetilde{P}$ is sent to the holonomy translate $\rho(g_j)P$.

The developed image of U is obtained by overlapping the five corner neighborhoods of $\rho(g_j)P_j$ in the holonomy-translated pentagons $\rho(g_j)P$.

Since the link of x has total angle 3π, the positive-real developed direction has exactly two lifts. For each of these two lifts, we identify the corresponding copy $g_j\widetilde{P}$ and use the holonomy map $\rho(g_j)$ to determine where a straight segment of developed length 1 ends. It remains to check whether either continuation terminates at the prescribed singular lift $\partial_0^{-1}ab\widetilde{s}_0$.

corner	incident lifted cut disk	angle interval
P_0	$\partial_0^{-1}b\widetilde{P}$	$[0, 3\pi/4]$
P_3	$\partial_0^{-1}ba^{-1}\widetilde{P}$	$[-\pi/2, 0]$
P_2	$\partial_0^{-1}ba^{-1}b^{-1}\widetilde{P}$	$[-\pi, -\pi/2]$
P_1	$\partial_0^{-1}ba^{-1}b^{-1}a\widetilde{P}$	$[-3\pi/2, -\pi]$
P_4	$\partial_0^{-1}b\partial_0\widetilde{P}$	$[-9\pi/4, -3\pi/2]$.

(50)

The order follows from the side identifications $A(P_0P_1) = P_3P_2, B(P_4P_3) = P_1P_2$, and the interval lengths are the five corner angles. The positive-real direction has exactly two preimages in $[-9\pi/4, 3\pi/4]$ which are 0 and -2π.

At angle 0, the arc follows the P_0P_1-edge of $\partial_0^{-1}b\widetilde{D}$, so its developed image is $[Q_1, Q_2]$ by holonomy trnaslated by $\rho(\partial_0^{-1}b)$. But its endpoint is

$$\partial_0^{-1}bw_1\widetilde{s}_0 = \partial_0^{-1}ba^{-1}ba\widetilde{s}_0 \neq \partial_0^{-1}ab\widetilde{s}_0 = x,$$

At angle -2π, the germ enters the interior of $d^{-1}bd\widetilde{D}$ from its P_4-corner. The linear part of $\rho(d^{-1}bd)$ is 2, so developed length 1 corresponds to base length 1/2, so before the lift reach the singularity, the developing image already reaches Q_2.

□

We are going to show $X_m \neq X_n$ generally.

Proposition 6.2 (Direct wrong-sheet obstruction for every central power)**.** *For every nonzero $m \in \mathbb{Z}$, the saddle connection prescribed by Eq. (78) does not exist on X_0. Consequently,*

$$X_m := Z^{-m}\Theta_{\kappa_0}(Z^m q_0) \neq X_0 \qquad (m \neq 0).$$

The proof of Proposition 6.2 is given in Appendix A.5.

Lemma 6.3 (The two-step room relation)**.** *Let*

$$\widetilde{C} := \sigma_1\sigma_2^{-1}\sigma_1^{-1}, \qquad K := Z^2\widetilde{C}^{-1}.$$

Then

$$X_{r+2} = K^{-1} \cdot X_r \qquad (r \in \mathbb{Z}).$$

Proof. By the room-coordinate formulas,

$$\widetilde{C}q_0 = [3 - 2i,\ 2 - i; \log 2, \log 2] = Z^2 q_0.$$

Moreover, $\widetilde{C}$ is the weak root slide, so

$$\widetilde{C} \cdot \Theta_{\kappa_0}(Z^r q_0) = \Theta_{\kappa_0}(\widetilde{C} \cdot Z^r q_0) = \Theta_{\kappa_0}(Z^{r+2} q_0).$$

Therefore, since $K^{-1} = Z^{-2}\widetilde{C}$,

$$X_{r+2} = Z^{-r-2}\Theta_{\kappa_0}(Z^{r+2}q_0) = Z^{-r-2}\widetilde{C} \cdot \Theta_{\kappa_0}(Z^r q_0) = K^{-1} \cdot X_r.$$

□

Corollary 6.4 (Pairwise separation at the symmetric parameter)**.** *One has $X_m \neq X_n$ whenever $m \neq n$.*

Proof. Suppose $X_m = X_n$. If $m \equiv n \pmod 2$, write $n = m + 2k$. Then **??** gives $X_n = K^{-k}X_m$. By Lemma 3.4, this forces $k = 0$.

If $m \not\equiv n \pmod 2$, one of the two indices is even; after swapping them, assume it is m. Applying $K^{m/2}$ to the equality gives

$$X_0 = X_{n-m}.$$

Here $n - m \neq 0$, contradicting Proposition 6.2. □

7 $\widehat{\mathrm{Hol}}^{-1}(\eta) = \{X_{Z^m\gamma_0}(\eta) \mid m \in \mathbb{Z}\}$

Let $\eta \in \mathcal{E}^{\dagger}$. Whenever $\gamma \in B_3$ satisfies $\gamma \cdot \eta \in \mathcal{E}_{++}$, put

$$q_\gamma := \Phi^{-1}(\gamma \cdot \eta) \in \mathcal{Y}_{++}$$

and define

$$X_\gamma(\eta) := \gamma^{-1} \cdot \Theta_{\kappa_0}(q_\gamma). \tag{51}$$

Then

$$\widehat{\mathrm{Hol}}(X_\gamma(\eta)) = \gamma^{-1} \cdot (\gamma \cdot \eta) = \eta. \tag{52}$$

The purpose of this section is to show that, after choosing one $\gamma_0 \in B_3$ with $\gamma_0 \cdot \eta \in \mathcal{E}_{++}$, every point in the fibre is represented by a central index:

$$\widehat{\mathrm{Hol}}^{-1}(\eta) = \{X_{Z^m\gamma_0}(\eta) \mid m \in \mathbb{Z}\}. \tag{53}$$

Lemma 7.1 (Existence of a positive presentation)**.** *For every $\eta \in \mathcal{E}^{\dagger}$, there exists $\gamma \in B_3$ such that*

$$\gamma \cdot \eta \in \mathcal{E}_{++}.$$

Equivalently, for every nonzero $\ell \in \mathbb{R}^2$, there exists $\gamma \in B_3$ such that

$$\Lambda(\gamma)\ell \in \mathbb{R}^2_{>0}.$$

Proof. Since $\Lambda : B_3 \to SL_2(\mathbb{Z})$ is surjective, write $\ell(\rho) = \begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} \neq 0$, we only need construct $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ such that

$$M\ell(\rho) = \begin{pmatrix} a\ell_1 + b\ell_2 \\ c\ell_1 + d\ell_2 \end{pmatrix} \in \mathbb{R}^2_{>0}.$$

First choose relatively prime integers a, b satisfying

$$a\ell_1 + b\ell_2 > 0.$$

For example, if $\ell_1 \neq 0$, take $(a, b) = (\operatorname{sign} \ell_1, 0)$, if $\ell_1 = 0$, take $(a, b) = (0, \operatorname{sign} \ell_2)$. Then we only need $N \in \mathbb{Z}$ big enough

$$c\ell_1 + d\ell_2 = (c_0\ell_1 + d_0\ell_2) + N(a\ell_1 + b\ell_2).$$

and need $ad - bc = a(d_0 + Nb) - b(c_0 + Na) = ad_0 - bc_0 = 1$.

Since $\gcd(a, b) = 1$, Bézout's identity gives these integers c_0, d_0. □

Corollary 7.2 (Positive edges are strict transfers)**.** *Let*

$$q = [e_1, e_2; \ell_1, \ell_2] \in \mathcal{Y}_{++}, \qquad s \in \{\sigma_1^{\pm 1}, \sigma_2^{\pm 1}\}.$$

Then

$$s \cdot q \in \mathcal{Y}_{++} \quad \Longleftrightarrow \quad \Lambda(s) \begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} \in \mathbb{R}^2_{>0}.$$

Proof. The generator formulas show that the new ordered basis has determinant

$$e^{\pm \ell_1} \omega(e_1, e_2) \qquad \text{or} \qquad e^{\pm \ell_2} \omega(e_1, e_2),$$

hence remains positively oriented. Its logarithmic coordinates are

$$\Lambda(s) \begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix}.$$

The conclusion now follows from **??**. □

Proposition 7.3 (Positive branch formula)**.** *Fix $\eta \in \mathcal{E}^\dagger$. For every $\gamma \in B_3$ with $\gamma\eta \in \mathcal{E}_{++}$, set*

$$q_\gamma := \Phi^{-1}(\gamma\eta), \qquad X_\gamma(\eta) := \gamma^{-1} \Theta_{\kappa_0}(q_\gamma).$$

Then

$$\widehat{\mathrm{Hol}}^{-1}(\eta) = \{X_\gamma(\eta) \mid \gamma\eta \in \mathcal{E}_{++}\}.$$

Different γ may represent the same point at this stage.

Proof. For every γ with $\gamma\eta \in \mathcal{E}_{++}$, $\widehat{\mathrm{Hol}}(X_\gamma(\eta)) = \gamma^{-1}\Phi(q_\gamma) = \eta$, so every $X_\gamma(\eta)$ lies in the fibre.

Conversely, let

$$X \in \widehat{\mathrm{Hol}}^{-1}(\eta).$$

By Proposition 5.3, $X \in \mathcal{U}_\kappa$ for some labelled cut system κ. By Lemma 4.6, write $\kappa = \gamma^{-1}\kappa_0$ for some $\gamma \in B_3$.

Suppose $X = \Theta_{\gamma^{-1}\kappa_0}(q)$ for the unique $q \in \mathcal{Y}_{++}$, Lemma 4.11 gives

$$\gamma \cdot X = \Theta_{\kappa_0}(q).$$

Hence we have

$$q = \Phi^{-1}(\widehat{\mathrm{Hol}}(\gamma \cdot X)) = \Phi^{-1}(\gamma\eta) = q_\gamma.$$

□

Fix $0 \neq \ell \in \mathbb{R}^2$. The positive-basis graph $\mathscr{G}_\ell$ has vertex set

$$\mathrm{Vert}(\mathscr{G}_\ell) = \left\{ M \in SL_2(\mathbb{Z}) \;\middle|\; M\ell \in \mathbb{R}^2_{>0} \right\}. \tag{54}$$

Two vertices are joined by a **strict-transfer edge** if they differ by left multiplication by $\Lambda(\sigma_1)^{\pm 1}$ or $\Lambda(\sigma_2)^{\pm 1}$, with both endpoints positive.

If ℓ has rational direction, a vertex with

$$M\ell = \begin{pmatrix} t \\ t \end{pmatrix}, \qquad t > 0,$$

is called a root. Consecutive roots are joined by the **weak root slide**

$$\widetilde{C} := \sigma_1 \sigma_2^{-1} \sigma_1^{-1}.$$

Theorem 7.4 (Positive-presentation reduction). *Let $\ell \in \mathbb{R}^2 \setminus \{0\}$. If ℓ has irrational direction, then $\mathscr{G}_\ell$ is connected. If ℓ has rational direction, then it becomes connected after adjoining the weak walks between consecutive roots.*

Hence, for any two vertices M_1, M_2, there exists a geometric word

$$\sigma = \sigma_r \cdots \sigma_1 \in B_3,$$

where each σ_i is a strict or weak walk, such that

$$M_2 = \Lambda(\sigma) M_1.$$

If $M_i = \Lambda(\gamma_i)$, then

$$X_{\sigma\gamma_1}(\eta) = X_{\gamma_1}(\eta).$$

Proof. The irrational case is Lemma A.2. In the rational case, Lemma A.4 gives the rooted components and the weak walks between consecutive roots. Branch constancy along strict and weak walks follows from **??** and Lemma A.6. $\square$

Theorem 7.5 (Reduction to central indices). *Let $\eta \in \mathcal{E}^\dagger$, and suppose that*

$$\gamma_1 \cdot \eta, \;\; \gamma_2 \cdot \eta \in \mathcal{E}_{++}.$$

Then there exists $m \in \mathbb{Z}$ such that

$$X_{\gamma_2}(\eta) = X_{Z^m\gamma_1}(\eta). \tag{55}$$

Proof. Put

$$M_i = \Lambda(\gamma_i).$$

By Theorem 7.4, there exists a geometric word $\sigma \in B_3$ such that

$$M_2 = \Lambda(\sigma) M_1 = \Lambda(\sigma\gamma_1).$$

Hence

$$\gamma_2(\sigma\gamma_1)^{-1} \in \ker \Lambda = \langle Z \rangle,$$

so, since Z is central,

$$\gamma_2 = Z^m \sigma \gamma_1 = \sigma Z^m \gamma_1$$

for some $m \in \mathbb{Z}$.

Since Z does not change the logarithmic holonomy, the same walk σ remains admissible. Therefore

$$X_{\gamma_2}(\eta) = X_{\sigma Z^m \gamma_1}(\eta) = X_{Z^m\gamma_1}(\eta).$$

$\square$

Theorem 7.6 (Central exhaustion). *Let $\eta \in \mathcal{E}^\dagger$, and choose $\gamma_0 \in B_3$ such that $\gamma_0 \cdot \eta \in \mathcal{E}_{++}$. Then*

$$\widehat{\mathrm{Hol}}^{-1}(\eta) = \{X_{Z^m\gamma_0}(\eta) \mid m \in \mathbb{Z}\}. \tag{56}$$

Repetitions are not yet excluded.

Proof. By Proposition 7.3, every fibre point is $X_\gamma(\eta)$ for some positive index γ. By Theorem 7.5,

$$X_\gamma(\eta) = X_{Z^m\gamma_0}(\eta)$$

for some $m \in \mathbb{Z}$.

Conversely, since $Z \in \ker \Lambda$, every $Z^m\gamma_0$ is again a positive index. Hence every point above belongs to the fibre. $\square$

8 $X_m(q) \neq X_n(q)$

We now prove that for every $q \in \mathcal{Y}_{++}$ and every $m \neq n$,

$$X_m(q) \neq X_n(q).$$

For $m, n \in \mathbb{Z}$, define the coincidence locus

$$A_{m,n} := \{q \in \mathcal{Y}_{++} \mid X_m(q) = X_n(q)\}. \tag{57}$$

Lemma 8.1 (Coincidence loci are clopen). *For every $m, n \in \mathbb{Z}$, the set $A_{m,n}$ is open and closed in $\mathcal{Y}_{++}$.*

Proof. Let $q_* \in A_{m,n}$, and put

$$X_* := X_m(q_*) = X_n(q_*).$$

By Proposition 5.3, $\widehat{\mathrm{Hol}}$ is a local diffeomorphism. Hence there is an open neighbourhood W of X_* on which $\widehat{\mathrm{Hol}}$ is injective.

Since X_m and X_n are continuous, there is a neighbourhood V of q_* such that

$$X_m(V) \cup X_n(V) \subset W.$$

For every $q \in V$,

$$\widehat{\mathrm{Hol}}(X_m(q)) = \Phi(q) = \widehat{\mathrm{Hol}}(X_n(q)).$$

Injectivity of $\widehat{\mathrm{Hol}}$ on W therefore gives $X_m(q) = X_n(q)$. Thus $A_{m,n}$ is open.

It remains to show that $A_{m,n}$ is closed. Let

$$\Delta \subset \mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \times \mathcal{T}^{\dagger}_{\mathrm{poly}}(S)$$

be the diagonal. Since $\mathcal{T}^{\dagger}_{\mathrm{poly}}(S)$ is Hausdorff, Δ is closed. Now

$$A_{m,n} = (X_m, X_n)^{-1}(\Delta),$$

and the map

$$(X_m, X_n) : \mathcal{Y}_{++} \longrightarrow \mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \times \mathcal{T}^{\dagger}_{\mathrm{poly}}(S)$$

is continuous. Hence $A_{m,n}$ is closed. □

Theorem 8.2 (No central repetitions). *For every $q \in \mathcal{Y}_{++}$ and every $m \neq n$,*

$$X_m(q) \neq X_n(q).$$

Proof. Since

$$\mathcal{Y}_{++} = \mathcal{B} \times \mathbb{R}^2_{>0}$$

is connected, every clopen set $A_{m,n}$ is either empty or all of $\mathcal{Y}_{++}$.

For $m \neq n$, the explicit computation at q_0 in Section 6 gives

$$X_m(q_0) \neq X_n(q_0).$$

Hence $q_0 \notin A_{m,n}$, so $A_{m,n} \neq \mathcal{Y}_{++}$. Therefore

$$A_{m,n} = \varnothing.$$

□

Corollary 8.3 (Exact classification of every fibre). *Let $\eta \in \mathcal{E}^{\dagger}$, and choose $\gamma_0 \in B_3$ such that $\gamma_0 \cdot \eta \in \mathcal{E}_{++}$. Then*

$$\widehat{\mathrm{Hol}}^{-1}(\eta) = \{X_{Z^m \gamma_0}(\eta) \mid m \in \mathbb{Z}\}, \tag{58}$$

Proof. Directly by Theorem 7.6 and 8.2.

□

9 The covering theorem

Theorem 9.1 (Global covering). *1. The enhanced holonomy map*

$$\widehat{\mathrm{Hol}} : \mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \longrightarrow \mathcal{E}^{\dagger}$$

is a covering map.

2. Over the positive chamber,

$$\widehat{\mathrm{Hol}}^{-1}(\mathcal{E}_{++}) = \coprod_{m\in\mathbb{Z}} X_m(\mathcal{Y}_{++}), \tag{59}$$

and each sheet maps diffeomorphically onto $\mathcal{E}_{++}$.

Proof. For (2), by 8.3,

$$\widehat{\mathrm{Hol}}^{-1}(\mathcal{E}_{++}) = \bigcup_{m\in\mathbb{Z}} X_m(\mathcal{Y}_{++}).$$

If $X_m(q) = X_n(q')$, then applying enhanced holonomy gives

$$\Phi(q) = \Phi(q'),$$

hence $q = q'$. By Theorem 8.2, $m = n$. Thus the union is disjoint.

Moreover,

$$\widehat{\mathrm{Hol}} \circ X_m = \Phi,$$

and

$$\Phi : \mathcal{Y}_{++} \longrightarrow \mathcal{E}_{++}$$

is a diffeomorphism. Since $X_m : \mathcal{Y}_{++} \to X_m(\mathcal{Y}_{++})$ is a diffeomorphism onto its image, it follows that

$$\widehat{\mathrm{Hol}}\big|_{X_m(\mathcal{Y}_{++})} : X_m(\mathcal{Y}_{++}) \longrightarrow \mathcal{E}_{++}$$

is a diffeomorphism.

Now let $\eta \in \mathcal{E}^{\dagger}$. Choose $\gamma \in B_3$ such that

$$\gamma \cdot \eta \in \mathcal{E}_{++},$$

and set

$$U := \gamma^{-1} \cdot \mathcal{E}_{++}.$$

Then U is an open neighbourhood of η. By equivariance,

$$\widehat{\mathrm{Hol}}^{-1}(U) = \gamma^{-1} \cdot \widehat{\mathrm{Hol}}^{-1}(\mathcal{E}_{++}).$$

Using part (2), we obtain

$$\widehat{\mathrm{Hol}}^{-1}(U) = \coprod_{m\in\mathbb{Z}} \gamma^{-1} \cdot X_m(\mathcal{Y}_{++}).$$

Moreover, by equivariance and part (2), for every m,

$$\widehat{\mathrm{Hol}}\big|_{\gamma^{-1}\cdot X_m(\mathcal{Y}_{++})} : \gamma^{-1} \cdot X_m(\mathcal{Y}_{++}) \longrightarrow U$$

is a diffeomorphism. Hence U is evenly covered. Since $\eta \in \mathcal{E}^{\dagger}$ was arbitrary, $\widehat{\mathrm{Hol}}$ is a covering map. $\square$

Proof. We first prove (2). By Corollary 8.3,

$$\widehat{\mathrm{Hol}}^{-1}(\mathcal{E}_{++}) = \bigcup_{m\in\mathbb{Z}} X_m(\mathcal{Y}_{++}).$$

If $X_m(q) = X_n(q')$, then

$$\Phi(q) = \widehat{\mathrm{Hol}}(X_m(q)) = \widehat{\mathrm{Hol}}(X_n(q')) = \Phi(q'),$$

so $q = q'$. By Theorem 8.2, $m = n$. Hence the union is disjoint.

Since $Z \in \ker \Lambda$, the action of Z preserves $\mathcal{Y}_{++}$. Therefore

$$X_m(\mathcal{Y}_{++}) = Z^{-m} \cdot \Theta_{\kappa_0}(\mathcal{Y}_{++}) = Z^{-m} \cdot \mathcal{U}_{\kappa_0}.$$

Thus each $X_m(\mathcal{Y}_{++})$ is open. Moreover,

$$\widehat{\mathrm{Hol}} \circ X_m = \Phi,$$

and $\Phi : \mathcal{Y}_{++} \to \mathcal{E}_{++}$ is a diffeomorphism. Hence

$$\widehat{\mathrm{Hol}}|_{X_m(\mathcal{Y}_{++})} : X_m(\mathcal{Y}_{++}) \longrightarrow \mathcal{E}_{++}$$

is a diffeomorphism. This proves (2).

We now prove (1). Let $\eta \in \mathcal{E}^{\dagger}$. By Lemma 7.1, choose $\gamma \in B_3$ such that

$$\gamma \cdot \eta \in \mathcal{E}_{++},$$

then

$$\eta \in= \gamma^{-1} \cdot \mathcal{E}_{++}.$$

By equivariance,

$$\widehat{\mathrm{Hol}}^{-1}(\gamma^{-1} \cdot \mathcal{E}_{++}) = \gamma^{-1} \cdot \widehat{\mathrm{Hol}}^{-1}(\mathcal{E}_{++}) = \coprod_{m\in\mathbb{Z}} \gamma^{-1} \cdot X_m(\mathcal{Y}_{++}).$$

Each set on the right is open, and by equivariance

$$\widehat{\mathrm{Hol}}|_{\gamma^{-1}\cdot X_m(\mathcal{Y}_{++})} : \gamma^{-1} \cdot X_m(\mathcal{Y}_{++}) \longrightarrow= \gamma^{-1} \cdot \mathcal{E}_{++}$$

is a diffeomorphism. Hence $\gamma^{-1} \cdot \mathcal{E}_{++}$ is evenly covered.

Since every $\eta \in \mathcal{E}^{\dagger}$ has such an open neighbourhood, $\widehat{\mathrm{Hol}}$ is a covering map. □

10 The lift monodromy and global coordinates

By Theorem 9.1, the enhanced holonomy map

$$\widehat{\mathrm{Hol}} : \mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \longrightarrow \mathcal{E}^{\dagger}$$

is a covering map. By Corollary 8.3, every fibre is indexed by $\mathbb{Z}$.

We now compute the monodromy of this covering and use it to identify the total space. Under the room coordinates,

$$\Phi : \mathcal{B} \times (\mathbb{R}^2 \setminus \{0\}) \xrightarrow{\sim} \mathcal{E}^{\dagger}.$$

10.1 The lifted B_3 action on the lifted logarithmic angle

Consider the universal covering

$$\varpi : \mathbb{R}^2 \longrightarrow \mathbb{R}^2 \setminus \{0\}, \qquad \varpi(s,\theta) = e^s \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \tag{60}$$

with deck transformations

$$\mathfrak{d}_k(s,\theta) = (s, \theta + 2\pi k), \qquad k \in \mathbb{Z}.$$

Recall

$$P := \Lambda(\sigma_1) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad Q := \Lambda(\sigma_2) = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Choose the paths

$$P_t = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, \qquad Q_t = \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix}, \qquad 0 \le t \le 1,$$

from the identity to P and Q, respectively.

Lifting the paths

$$t \longmapsto P_t \varpi(s,\theta), \qquad t \longmapsto Q_t \varpi(s,\theta)$$

through ϖ, starting at (s,θ), defines lifts

$$\widetilde{P}, \widetilde{Q} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2.$$

Consider the lift

$$\widetilde{\Lambda} : B_3 \longrightarrow \widetilde{SL_2(\mathbb{R})}, \qquad \widetilde{\Lambda}(\sigma_1) = \widetilde{P}, \quad \widetilde{\Lambda}(\sigma_2) = \widetilde{Q},$$

with

$$\varpi \circ \widetilde{\Lambda}(\gamma) = \Lambda(\gamma) \circ \varpi \qquad (\gamma \in B_3). \tag{61}$$

Moreover, $\varpi \circ \widetilde{\Lambda}(\gamma) \circ \mathfrak{d}_k = \Lambda(\gamma) \circ \varpi \circ \mathfrak{d}_k = \Lambda(\gamma) \circ \varpi = \varpi \circ \widetilde{\Lambda}(\gamma) = \varpi \circ \mathfrak{d}_k \circ \widetilde{\Lambda}(\gamma)$, gives

$$\widetilde{\Lambda}(\gamma) \circ \mathfrak{d}_k = \mathfrak{d}_k \circ \widetilde{\Lambda}(\gamma) \qquad (\gamma \in B_3, \ k \in \mathbb{Z}). \tag{62}$$

Since $\widetilde{\Lambda}(\gamma)$ maps $\varpi^{-1}(\ell)$ bijectively onto $\varpi^{-1}(\Lambda(\gamma)\ell)$,

Lemma 10.1 (Central winding)**.** *We have*

$$\widetilde{\Lambda}(Z)(s,\theta) = (s, \theta + 2\pi).$$

Proof. Since $\Lambda(Z) = (QP)^6 = I$, Eq. (61) gives $\varpi \circ \widetilde{\Lambda}(Z) = \varpi$.

Hence $\widetilde{\Lambda}(Z) = \mathfrak{d}_k$ for some $k \in \mathbb{Z}$.

It remains to determine k. Take

$$\ell_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \varpi\left(\log\sqrt{2}, \frac{\pi}{4}\right).$$

Following the twelve shear paths in $Z = (\sigma_2\sigma_1)^6$, the successive endpoints are

$$\begin{gathered} (1,1),\ (1,2),\ (-1,2),\ (-1,1),\ (-2,1), \\ (-2,-1),\ (-1,-1),\ (-1,-2),\ (1,-2), \\ (1,-1),\ (2,-1),\ (2,1),\ (1,1). \end{gathered}$$

Each shear piece is the straight segment between two consecutive points, and none of these segments contains the origin. Following the continuous argument along the twelve segments gives

$$\frac{\pi}{4} \longmapsto \frac{\pi}{4} + 2\pi.$$

Thus the path winds once counterclockwise around the origin, so $k = 1$. $\square$

10.2 Lifted angles of positive presentations

For $\Lambda(\gamma)\ell \in \mathbb{R}^2_{>0}$, there are unique $r > 0$ and $0 < \alpha < \frac{\pi}{2}$ such that

$$\Lambda(\gamma)\ell = r\begin{pmatrix}\cos\alpha\\ \sin\alpha\end{pmatrix}.$$

Then the lift could write as

$$\widetilde{\Lambda}(\gamma)\varpi^{-1}(\ell) = \varpi^{-1}(\Lambda(\gamma)\ell) = \{(\log r, \alpha + 2\pi k) \mid k \in \mathbb{Z}\},$$

and have exactly one of these points lies in

$$\mathbb{R} \times \left(0, \frac{\pi}{2}\right).$$

we then define $\widetilde{\ell}_\gamma \in \varpi^{-1}(\ell)$ to be the unique point such that

$$\widetilde{\Lambda}(\gamma)(\widetilde{\ell}_\gamma) \in \mathbb{R} \times \left(0, \frac{\pi}{2}\right). \tag{63}$$

Lemma 10.2 (Compatibility of lifted angles)**.** *Let* $0 \neq \ell \in \mathbb{R}^2$*. Recall* $\mathscr{P}_\ell := \left\{\gamma \in B_3 \mid \Lambda(\gamma)\ell \in \mathbb{R}^2_{>0}\right\}$.

1. If $\gamma, \gamma' \in \mathscr{P}_\ell$ *, we know they are joined by one strict or weak walk, and we now have*

$$\widetilde{\ell}_{\gamma'} = \widetilde{\ell}_\gamma.$$

2. If $\gamma \in \mathscr{P}_\ell$*, then for every* $m \in \mathbb{Z}$*, we know* $Z^m\gamma \in \mathscr{P}_\ell$ *and we now have*

$$\widetilde{\ell}_{Z^m\gamma} = \mathfrak{d}_{-m}(\widetilde{\ell}_\gamma), \tag{64}$$

where

$$\mathfrak{d}_k : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \qquad \mathfrak{d}_k(s,\theta) = (s, \theta + 2\pi k).$$

Proof. For (1), For (1), suppose first that $\gamma' = s\gamma$, with $s \in \{\sigma_1^{\pm 1}, \sigma_2^{\pm 1}\}$, is one strict step. If $\Lambda(\gamma)\ell = (x,y)^\mathsf{T} \in \mathbb{R}^2_{>0}$, then the corresponding shear path is one of

$$\begin{pmatrix}x\\ y \pm tx\end{pmatrix}, \qquad \begin{pmatrix}x \pm ty\\ y\end{pmatrix}, \qquad 0 \le t \le 1.$$

Since the step is strict, this path remains in $\mathbb{R}^2_{>0}$, and hence makes no full turn around the origin.

For one weak walk at a rational root, the logarithmic vector follows

$$\begin{pmatrix}c\\ c\end{pmatrix} \longmapsto \begin{pmatrix}c\\ 0\end{pmatrix} \longmapsto \begin{pmatrix}c\\ 0\end{pmatrix} \longmapsto \begin{pmatrix}c\\ c\end{pmatrix}, \qquad c > 0,$$

which also makes no full turn around the origin.

Since $\gamma' = s\gamma$, we have

$$\widetilde{\Lambda}(\gamma') = \widetilde{\Lambda}(s)\widetilde{\Lambda}(\gamma).$$

By the definition of $\widetilde{\ell}_\gamma$,

$$\widetilde{\Lambda}(\gamma)(\widetilde{\ell}_\gamma) \in \mathbb{R} \times \left(0, \frac{\pi}{2}\right).$$

The lifted shear $\widetilde{\Lambda}(s)$ follows the corresponding shear path, which remains in $\mathbb{R}^2_{>0}$. Hence

$$\widetilde{\Lambda}(\gamma')(\widetilde{\ell}_\gamma) = \widetilde{\Lambda}(s)\widetilde{\Lambda}(\gamma)(\widetilde{\ell}_\gamma) \in \mathbb{R} \times \left(0, \frac{\pi}{2}\right). \tag{65}$$

By uniqueness in Eq. (63), we know $\widetilde{\ell}_{\gamma'}$ is the only point satisfying 65 so we have

$$\widetilde{\ell}_{\gamma'} = \widetilde{\ell}_{\gamma}.$$

For (2), Since $\varpi \circ \mathfrak{d}_{-m} = \varpi$, for $\widetilde{\ell}_{\gamma} \in \varpi^{-1}(\ell)$ we have

$$\varpi\Big(\mathfrak{d}_{-m}(\widetilde{\ell}_{\gamma})\Big) = \varpi(\widetilde{\ell}_{\gamma}) = \ell \Longrightarrow \mathfrak{d}_{-m}(\widetilde{\ell}_{\gamma}) \in \varpi^{-1}(\ell).$$

Furthermore, since $\widetilde{\Lambda}(Z^m) = \mathfrak{d}_m$, we have

$$\left.\begin{aligned}\widetilde{\Lambda}(Z^m\gamma)\left(\mathfrak{d}_{-m}(\widetilde{\ell}_{\gamma})\right) &= \widetilde{\Lambda}(Z^m)\widetilde{\Lambda}(\gamma)\left(\mathfrak{d}_{-m}(\widetilde{\ell}_{\gamma})\right)\\ &= \mathfrak{d}_m\widetilde{\Lambda}(\gamma)\left(\mathfrak{d}_{-m}(\widetilde{\ell}_{\gamma})\right)\\ &= \mathfrak{d}_m\mathfrak{d}_{-m}\widetilde{\Lambda}(\gamma)(\widetilde{\ell}_{\gamma})\\ &= \widetilde{\Lambda}(\gamma)(\widetilde{\ell}_{\gamma}) \in \mathbb{R} \times \left(0, \frac{\pi}{2}\right).\end{aligned}\right. \Longrightarrow \widetilde{\ell}_{Z^m\gamma} = \mathfrak{d}_{-m}(\widetilde{\ell}_{\gamma}).$$

□

10.3 Global coordinates and monodromy

Consider two covering map

$$H := \Phi^{-1} \circ \widehat{\mathrm{Hol}} : \mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \longrightarrow \mathcal{B} \times (\mathbb{R}^2 \setminus \{0\}).$$

and

$$p := \mathrm{id}_{\mathcal{B}} \times \varpi : \mathcal{B} \times \mathbb{R}^2 \longrightarrow \mathcal{B} \times (\mathbb{R}^2 \setminus \{0\}), \qquad \varpi(s,\theta) = e^s(\cos\theta, \sin\theta).$$

For a presentation $X_\gamma(\eta)$, write $(\beta, \ell) := \Phi^{-1}(\eta)$. We call it positive if $\Lambda(\gamma)\ell \in \mathbb{R}^2_{>0}$. For such a positive presentation, let $\widetilde{\ell}_{\gamma} \in \varpi^{-1}(\ell) \subset \mathbb{R}^2$ be the unique lift satisfying $\widetilde{\Lambda}(\gamma)(\widetilde{\ell}_{\gamma}) \in \mathbb{R} \times (0, \pi/2)$.

Define

$$\widetilde{H} : \mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \longrightarrow \mathcal{B} \times \mathbb{R}^2, \qquad \widetilde{H}(X_\gamma(\eta)) := (\beta, \widetilde{\ell}_{\gamma}). \tag{66}$$

Since $\varpi(\widetilde{\ell}_{\gamma}) = \ell$, we have

$$p(\widetilde{H}(X_\gamma(\eta))) = (\mathrm{id}_{\mathcal{B}} \times \varpi)(\beta, \widetilde{\ell}_{\gamma}) = (\beta, \ell) = H(X_\gamma(\eta)).$$

Hence $p \circ \widetilde{H} = H$, so to identify the two coverings, we only need to prove that $\widetilde{H}$ is a homeomorphism:

Theorem 10.3 (Global coordinates)**.** *The map*

$$\widetilde{H} : \mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \longrightarrow \mathcal{B} \times \mathbb{R}^2, \qquad \widetilde{H}(X_\gamma(\eta)) := (\beta, \widetilde{\ell}_{\gamma}). \tag{67}$$

has the following properties.

(1) $\widetilde{H}$ *is well defined and continuous.*

(2) $\widetilde{H}$ *is fibre bijection for* $q = (\beta, \ell) \in \mathcal{Y}_{++}$.

$$\widetilde{H} : H^{-1}(q) \xrightarrow{\sim} p^{-1}(q),$$

More precisely, we have

$$\widetilde{H}(X_m(q)) = (\beta, \log|\ell|, \theta - 2\pi m), \qquad \theta = \arg \ell \in (0, \pi/2). \tag{68}$$

(3) $\widetilde{H}$ *is fibre bijection for any* $q \in \mathcal{B} \times (\mathbb{R}^2 \setminus \{0\})$*, so it is bijection and then diffeomorphism, so we have*

$$\mathcal{T}^{\dagger}_{\mathrm{poly}}(S) \cong \mathcal{B} \times \mathbb{R}^2 \cong SL_2(\mathbb{R}) \times \mathbb{R}^2.$$

Proof. (1) We need to show

$$X_\gamma(\eta) = X_{\gamma'}(\eta) \quad \Longrightarrow \quad \widetilde{H}(X_\gamma(\eta)) = (\beta, \widetilde{\ell}_\gamma) = (\beta, \widetilde{\ell}_{\gamma'}) = \widetilde{H}(X_{\gamma'}(\eta)).$$

Recall that, for a positive presentation δ,

$$X_\delta(\eta) := \delta^{-1} \cdot \Theta_{\kappa_0}(\delta \cdot (\beta, \ell)), \qquad (\beta, \ell) := \Phi^{-1}(\eta).$$

By the connectivity of the positive-presentation graph, and γ' are joined by strict and weak walks. Hence Lemma 10.2 gives $\widetilde{\ell}_\gamma = \widetilde{\ell}_{\gamma'}$.

(2) We already know

$$H^{-1}(q) = \{X_m(q) \mid m \in \mathbb{Z}\},$$

with all $X_m(q)$ distinct and

$$p^{-1}(q) = \{(\beta, \log|\ell|, \theta + 2\pi k) \mid k \in \mathbb{Z}\}.$$

We only need show $\widetilde{H}$ identifies the two fibres by $k = -m$. By definition $\widetilde{H}(X_\gamma(\eta)) := (\beta, \widetilde{\ell}_\gamma)$ we only need show that $\widetilde{\ell}_{Z^m} = (\log|\ell|, \theta - 2\pi m)$ then we get

$$\widetilde{H}(X_m(q)) = (\beta, \widetilde{\ell}_{Z^m}) = (\beta, \log|\ell|, \theta - 2\pi m).$$

Let $q = (\beta, \ell) \in \mathcal{Y}_{++}$ and write $\theta = \arg \ell \in (0, \pi/2)$. For the identity presentation $\gamma = e$, the selected lift $\widetilde{\ell}_e \in \varpi^{-1}(\ell)$ is characterized by

$$\widetilde{\Lambda}(e)(\widetilde{\ell}_e) \in \mathbb{R} \times \left(0, \frac{\pi}{2}\right).$$

Since $\widetilde{\Lambda}(e) = \mathrm{id}$, it follows that

$$\widetilde{\ell}_e = (\log|\ell|, \theta).$$

For the presentation Z^m, Eq. (64) gives

$$\widetilde{\ell}_{Z^m} = \mathfrak{d}_{-m}(\widetilde{\ell}_e) = (\log|\ell|, \theta - 2\pi m).$$

(3) We only need to prove that the diagram

$$\begin{tikzcd}
H^{-1}(q_0) \arrow[r, "\widetilde{H}", "\sim"'] \arrow[d, "T^H_c"'] & p^{-1}(q_0) \arrow[d, "T^p_c"] \\
H^{-1}(q) \arrow[r, "\widetilde{H}"'] & p^{-1}(q)
\end{tikzcd}$$

commutes, where for $X \in H^{-1}(q_0)$ and $z \in p^{-1}(q_0)$,

$$T^H_c(X) := \widetilde{c}^H_X(1), \qquad T^p_c(z) := \widetilde{c}^p_z(1),$$

and $\widetilde{c}^H_X$, $\widetilde{c}^p_z$ are the lifts of a path c from q_0 to q under H and p, respectively. Both T^H_c and T^p_c are bijections, since lifting the reversed path $\bar{c}$ gives their inverses.

For commutativity,

$$\widetilde{H}(T^H_c(X)) = \widetilde{H} \circ \widetilde{c}^H_X(1) = \widetilde{c}^p_{\widetilde{H}(X)}(1) = T^p_c(\widetilde{H}(X)).$$

let $X \in H^{-1}(q_0)$ we only need to show

$$\widetilde{H} \circ \widetilde{c}_X^H = \widetilde{c}^p_{\widetilde{H}(X)}.$$

since

$$p \circ \widetilde{H} \circ \widetilde{c}_X^H = H \circ \widetilde{c}_X^H = c.$$

the path $\widetilde{H} \circ \widetilde{c}_X^H$ is a lift of c under p with initial point $\widetilde{H}(X) \in p^{-1}(q_0)$, hence, by uniqueness of path lifting with the same initial point, we get $\widetilde{H} \circ \widetilde{c}_X^H = \widetilde{c}^p_{\widetilde{H}(X)}$.

It remains to show that the bijection $\widetilde{H}$ is a diffeomorphism, We only need to show that $\widetilde{H}$ is a local diffeomorphism.

Fix $X \in \mathcal{T}^\dagger_{\text{poly}}(S)$, and choose a connected neighbourhood U of $H(X)$ evenly covered by both H and p. Let V and W be the sheets over U containing X and $\widetilde{H}(X)$, respectively. Then

$$H|_V : V \xrightarrow{\sim} U, \qquad p|_W : W \xrightarrow{\sim} U$$

are diffeomorphisms. Then we only need to show

$$\widetilde{H}|_V = (p|_W)^{-1} \circ H|_V.$$

which only need $p|_W \circ \widetilde{H}|_V = H|_V$ which only need diagram commutative

$$\begin{tikzcd} V \arrow[rr, "\widetilde{H}|_V"] \arrow[dr, "H|_V"'] & & W \arrow[dl, "p|_W"] \\ & U & \end{tikzcd}$$

it remains only to check that $\widetilde{H}(V) \subset W$. Since V is connected, so is $\widetilde{H}(V)$. As

$$p^{-1}(U) = \coprod_\alpha W_\alpha$$

is the disjoint union of connected component and $\widetilde{H}(X) \in W$, we have $\widetilde{H}(V) \subset W$. $\square$

Fix

$$q_0 = (\beta_0, \ell_0) \in \mathcal{Y}_{++}, \qquad \theta_0 := \arg \ell_0, \qquad X_m := X_m(q_0).$$

Let R_θ denote counterclockwise rotation through angle θ. The loops

$$a(t) = (R_{2\pi t}\beta_0, \ell_0), \qquad b(t) = (\beta_0, R_{2\pi t}\ell_0), \qquad 0 \le t \le 1,$$

generate $\pi_1(\mathcal{B} \times (\mathbb{R}^2 \setminus \{0\}), q_0) = \langle a, b \mid ab = ba \rangle$. And give the monodromy action on the covering:

Corollary 10.4 (Angular monodromy)**.** *The monodromy action of $\pi_1(\mathcal{B} \times (\mathbb{R}^2 \setminus \{0\}), q_0) = \langle a, b \mid ab = ba \rangle$ on the fibre $H^{-1}(q_0) = \{X_m \mid m \in \mathbb{Z}\}$ is given by*

$$a \cdot X_m = X_m, \qquad b \cdot X_m = X_{m-1}.$$

Hence, for all $j, k, m \in \mathbb{Z}$,

$$a^j b^k \cdot X_m = X_{m-k}.$$

In particular,

$$H_* \pi_1(\mathcal{T}^\dagger_{\text{poly}}(S), X_0) = \langle a \rangle = \mathbb{Z} \times \{0\}.$$

Proof. By Eq. (68),
$$\widetilde{H}(X_m) = (\beta_0, \log|\ell_0|, \theta_0 - 2\pi m).$$
The lifts of a and b under p, starting at this point, are
$$t \longmapsto (R_{2\pi t}\beta_0, \log|\ell_0|, \theta_0 - 2\pi m), \qquad t \longmapsto (\beta_0, \log|\ell_0|, \theta_0 - 2\pi m + 2\pi t).$$
Their endpoints are
$$\widetilde{H}(X_m) \qquad \text{and} \qquad \widetilde{H}(X_{m-1}),$$
Since $\widetilde{H}$ is a covering isomorphism, this gives
$$a \cdot X_m = X_m, \qquad b \cdot X_m = X_{m-1}.$$
□

A The positive-basis forest and the rational root slide

This appendix proves the combinatorial and geometric assertions used in Theorem 7.4. Put

$$P := \Lambda(\sigma_1) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad Q := \Lambda(\sigma_2) = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Fix $0 \neq \ell \in \mathbb{R}^2$. The graph $\mathscr{G}_\ell$ has vertex set

$$\left\{ M \in SL_2(\mathbb{Z}) \;\middle|\; M\ell \in \mathbb{R}^2_{>0} \right\},$$

with unoriented edges between M and PM, and between M and $Q^{-1}M$, whenever both endpoints are positive.

A.1 The positive-basis forest

Lemma A.1 (Forest structure)**.** *Let $M \in \mathscr{G}_\ell$ and write*

$$M\ell = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad x, y > 0.$$

Then:

1. *adjacent vertices have different heights;*
2. *every vertex has at most one neighbour of smaller height. More precisely:*
   (a) *if $x > y$, then QM is the unique neighbour of smaller height;*
   (b) *if $y > x$, then $P^{-1}M$ is the unique neighbour of smaller height;*
   (c) *if $x = y$, then M has no neighbour of smaller height.*

In particular, $\mathscr{G}_\ell$ has no cycles. Hence every connected component of $\mathscr{G}_\ell$ is a tree.

Proof. The possible adjacent pairs are

$$PM\ell = (x, x+y), \qquad Q^{-1}M\ell = (x+y, y),$$
$$P^{-1}M\ell = (x, y-x), \qquad QM\ell = (x-y, y),$$

whenever the last two corresponding pair is positive. Define $h(M) := x + y$. then the possible adjacent heights are

$$h(PM) = h(M) + x, \qquad h(Q^{-1}M) = h(M) + y,$$
$$h(P^{-1}M) = h(M) - x, \qquad h(QM) = h(M) - y.$$

Hence adjacent vertices have different heights. If $x > y$, then $P^{-1}M$ is not positive, only QM is positive while having a smaller height. The case $y > x$ is analogous. If $x = y$, both QM and $P^{-1}M$ are not positive, so there is no smaller neighbour.

Now suppose that a cycle exists and choose a vertex to be the maximal height on the cycle. Since adjacent heights are different, its two neighbours on the cycle both have smaller height , contradicting the uniqueness above. Hence there is no cycle, so every connected component is a tree. $\square$

A.2 Irrational connectivity

Lemma A.2 (Irrational connectivity)**.** *If ℓ has irrational direction, then $\mathscr{G}_\ell$ is connected.*

Proof. **Step 1: $x \neq y$ at every vertex.** Assume that ℓ has irrational direction:

$$r\ell \neq 0 \qquad \text{for every } 0 \neq r \in \mathbb{Z}^{1\times 2}.$$

Fix a vertex $M \in \mathscr{G}_\ell$, write its rows as r_1, r_2, and put

$$x = r_1\ell, \qquad y = r_2\ell.$$

We claim that $x/y \notin \mathbb{Q}$ and in particular

$$x \neq y$$

Indeed, if $x/y = p/q \in \mathbb{Q}$, then

$$(qr_1 - pr_2)\ell = qx - py = 0.$$

Since r_1, r_2 form an integral basis, $qr_1 - pr_2 \neq 0$, contradicting the irrational direction of ℓ.

Step 2: The descending sequence. Since $x \neq y$ at every vertex, Lemma A.1 shows that every vertex has a unique neighbour of smaller height. Starting from $M = M_0$, we therefore obtain a unique descending path

$$M_0 \searrow M_1 \searrow M_2 \searrow \cdots, \qquad M_n = \begin{pmatrix} r_1^{(n)} \\ r_2^{(n)} \end{pmatrix}.$$

where

$$r_2^{(n)}\ell > r_1^{(n)}\ell \Longrightarrow M_{n+1} = \begin{pmatrix} r_1^{(n)} \\ r_2^{(n)} - r_1^{(n)} \end{pmatrix}, \qquad r_1^{(n)}\ell > r_2^{(n)}\ell \Longrightarrow M_{n+1} = \begin{pmatrix} r_1^{(n)} - r_2^{(n)} \\ r_2^{(n)} \end{pmatrix}. \tag{69}$$

Step 3: Connect. Let

$$M' = \begin{pmatrix} r_1' \\ r_2' \end{pmatrix}$$

be any other vertex. To show that M_n is connected to M', we wish to find a product of P and Q^{-1} , both of which preserve positivity, such a product has the form

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}), \qquad A \geq 0.$$

such that

$$M' = \begin{pmatrix} r_1' \\ r_2' \end{pmatrix} = \begin{pmatrix} ar_1^{(n)} + br_2^{(n)} \\ cr_1^{(n)} + dr_2^{(n)} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} r_1^{(n)} \\ r_2^{(n)} \end{pmatrix} = AM_n. \tag{70}$$

To find such an A, it is enough to show that every

$$r' \in H_\ell := \left\{ r \in \mathbb{R}^{1\times 2} \;\middle|\; r\ell > 0 \right\}$$

lies in

$$C_n := \mathbb{R}_{\geq 0} r_1^{(n)} + \mathbb{R}_{\geq 0} r_2^{(n)} \subset \mathbb{R}^{1\times 2}, \qquad r_i^{(n)}\ell > 0.$$

Where we have $C_n \subset C_{n+1}$ by (69).

Thus it remains to show that

$$H_\ell \subset \bigcup_{n\geq 0} C_n.$$

Since for every vertex M' we can find M_n and a nonnegative matrix A satisfying (70), we also have

$$A = M' M_n^{-1} \in SL_2(\mathbb{Z}).$$

By Lemma A.3, A is a product of P and Q^{-1}. Since both preserve positivity, this product gives a path in $\mathscr{G}_\ell$ from M_n to M'. Hence M' is connected to the descending path from M.

Since clearly $\bigcup_{n\geq 0} C_n \subset H_\ell \cup \{0\}$, we actually could get

$$\bigcup_{n\geq 0} C_n = H_\ell \cup \{0\}.$$

Step 4: proof of $H_\ell \subset \bigcup_{n\geq 0} C_n$. Write

$$\ell = \begin{pmatrix} \ell_1 \\ \ell_2 \end{pmatrix} \neq 0, \qquad e_1 = \begin{pmatrix} -\ell_2, \ell_1 \end{pmatrix}, \qquad e_2 = \begin{pmatrix} \ell_1, \ell_2 \end{pmatrix}.$$

Then every $r \in \mathbb{R}^{1\times 2}$ can be written uniquely as $r = se_1 + te_2$. Since $r\ell = t(\ell_1^2 + \ell_2^2)$, we have

$$H_\ell = \left\{ r \in \mathbb{R}^{1\times 2} \,\middle|\, r\ell > 0 \right\} = \{ se_1 + te_2 \mid s \in \mathbb{R},\ t > 0 \}.$$

Recall $C_n := \mathbb{R}_{\geq 0} r_1^{(n)} + \mathbb{R}_{\geq 0} r_2^{(n)} \subset \mathbb{R}^{1\times 2}$, where $r_i^{(n)} = s_i^{(n)} e_1 + t_i^{(n)} e_2 \in H_\ell$, if we set

$$\alpha_n := \min\left\{ \frac{s_1^{(n)}}{t_1^{(n)}}, \frac{s_2^{(n)}}{t_2^{(n)}} \right\}, \qquad \beta_n := \max\left\{ \frac{s_1^{(n)}}{t_1^{(n)}}, \frac{s_2^{(n)}}{t_2^{(n)}} \right\}.$$

then we have

$$C_n = \{0\} \cup \left\{ se_1 + te_2 \,\middle|\, t > 0,\ \alpha_n \leq \frac{s}{t} \leq \beta_n \right\}.$$

Since $C_n \subset C_{n+1}$, we have $[\alpha_n, \beta_n] \subset [\alpha_{n+1}, \beta_{n+1}]$, then $H_\ell \subset \bigcup_{n\geq 0} C_n$ is equivalent to

$$\alpha_n \longrightarrow -\infty, \qquad \beta_n \longrightarrow +\infty.$$

So we only need to show

$$\left| \frac{s_i^{(n)}}{t_i^{(n)}} \right| \longrightarrow \infty, \qquad i = 1, 2,$$

Since

$$r_i^{(n)} = s_i^{(n)} e_1 + t_i^{(n)} e_2, \qquad r_i^{(n)} \ell = t_i^{(n)} (\ell_1^2 + \ell_2^2),$$

the Euclidean algorithm for an irrational ratio gives

$$r_i^{(n)} \longrightarrow 0 \Longrightarrow t_i^{(n)} \longrightarrow 0.$$

On the other hand, since $e_1 \perp e_2$ and $\|e_1\| = \|e_2\| = \|\ell\|$, we have

$$\|r_i^{(n)}\|^2 = \|\ell\|^2 \left((s_i^{(n)})^2 + (t_i^{(n)})^2 \right) \to \infty.$$

and since we already have $t_i^{(n)} \longrightarrow 0$, hence

$$|s_i^{(n)}| \longrightarrow \infty \Longrightarrow \left| \frac{s_i^{(n)}}{t_i^{(n)}} \right| \longrightarrow \infty..$$

The reason of $\|r_i^{(n)}\| \to \infty$: if it is bounded, since $\mathbb{Z}^2$ is discrete, the bounded subsequence would admit a constant subsequence $r_i^{(n_k)} = r \neq 0$. Along this subsequence, $r\ell = r_i^{(n_k)} \ell \to 0$, hence $r\ell = 0$, contradicting the irrationality of ℓ.

$\square$

Lemma A.3 (Nonnegative matrices in $SL_2(\mathbb{Z})$)**.** *Let*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}), \qquad A \geq 0.$$

Then A is a product of P and Q^{-1}.

Proof. Suppose first that $A \neq I$. We claim that the two rows of A are one of the following:

$$a \leq c,\ b \leq d \qquad \text{or} \qquad c \leq a,\ d \leq b.$$

other wise we have

$$\begin{cases} a \leq c-1,\ b \geq d+1 \Longrightarrow 1 = ad - bc \leq (c-1)d - (d+1)c = -c - d < 0, \\ a \geq c+1,\ b \leq d-1 \Longrightarrow 1 = ad - bc \geq (c+1)d - (d-1)c = c + d \Longrightarrow (c,d) = (0,1) \Longrightarrow (a,b) = (1,0) \Longrightarrow A \end{cases}$$

Then we have

$$A \longmapsto A' = \begin{cases} P^{-1}A = \begin{pmatrix} a & b \\ c-a & d-b \end{pmatrix}, & a \leq c,\ b \leq d, \\ QA = \begin{pmatrix} a-c & b-d \\ c & d \end{pmatrix}, & c \leq a,\ d \leq b. \end{cases} \qquad A' \in SL_2(\mathbb{Z}), \qquad A' \geq 0,$$

and the sum of its entries is strictly smaller than that of A. Thus, after finitely many steps,

$$A = A_0 \longrightarrow A_1 \longrightarrow \cdots \longrightarrow A_k = I.$$

□

A.3 Rational roots

Suppose that

$$\ell = cv, \qquad c > 0,$$

where $v \in \mathbb{Z}^2$ is primitive. Choose $r_0, k \in \mathbb{Z}^{1\times 2}$ such that

$$r_0 v = 1, \qquad kv = 0, \qquad \det \begin{pmatrix} r_0 \\ k \end{pmatrix} = 1.$$

Lemma A.4 (Rational roots)**.** *Suppose that ℓ has rational direction. Then every connected component of $\mathscr{G}_\ell$ contains a unique root,*

$$R_n := \begin{pmatrix} r_0 + nk \\ r_0 + (n+1)k \end{pmatrix}, \qquad n \in \mathbb{Z}. \tag{71}$$

Moreover, put

$$\widetilde{C} := \sigma_1 \sigma_2^{-1} \sigma_1^{-1}, \qquad C_0 := \Lambda(\widetilde{C}) = PQ^{-1}P^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}.$$

Then

$$C_0 R_n = R_{n+1}.$$

Thus consecutive components are connected by the weak move represented by $\widetilde{C}$.

Proof. **Step 1: Every descending path terminates at $(1,1)$.** Since $M\ell > 0$ and $\ell = cv$, write

$$Mv = \begin{pmatrix} r_1 v \\ r_2 v \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}^2_{>0}.$$

Moreover, Mv is primitive. Otherwise, if $d \mid x$ and $d \mid y$, then $Mv = dw$ for some $w \in \mathbb{Z}^2$, and hence

$$v = M^{-1}(Mv) = d\,M^{-1}w.$$

Since v is primitive, $d = 1$. Thus

$$\gcd(x, y) = 1.$$

Since x and y are positive coprime integers, the the Euclidean algorithm process terminates at $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. We call M a root if $Mv = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Step 2: Classification of the roots.
Choose $r_0, k \in \mathbb{Z}^{1\times 2}$ such that

$$r_0 v = 1, \qquad kv = 0, \qquad \det \begin{pmatrix} r_0 \\ k \end{pmatrix} = 1.$$

Then every integral row r satisfying $rv = 1$ has the form

$$r = r_0 + nk, \qquad n \in \mathbb{Z}.$$

since

$$rv = r_0 v = 1 \Longrightarrow (r - r_0)v = 0 \Longrightarrow r - r_0 \in \ker(v) = \mathbb{Z}k.$$

Hence a root has the form $M = \begin{pmatrix} r_0 + nk \\ r_0 + mk \end{pmatrix}$ for some $m, n \in \mathbb{Z}$. Since

$$1 = \det M = \det \begin{pmatrix} r_0 + nk \\ r_0 + mk \end{pmatrix} = m - n,$$

we have $m = n + 1$, and therefore $M = R_n = \begin{pmatrix} r_0 + nk \\ r_0 + (n+1)k \end{pmatrix}$.

Step 3: Distinct roots lie in distinct components.
Suppose that R_n and R_m, with $n \neq m$, lie in the same connected component.
By Lemma A.1, this component is a tree, so there is a unique path

$$R_n = M_0, M_1, \ldots, M_N = R_m.$$

A root has no neighbour of smaller height, so

$$h(M_1) > h(R_n), \qquad h(M_{N-1}) > h(R_m).$$

Hence the vertex with maximal height on this path is an interior vertex, and its two neighbours on the path both have smaller height since any vertex has no neighbour of smaller height by Lemma A.1. This contradicts Lemma A.1, since every vertex has at most one neighbour of smaller height. □

A.4 Geometric realization of strict and weak walks

Lemma A.5 (Strict walk preserves the marked surface)**.** *Let $s \in \{\sigma_1^{\pm 1}, \sigma_2^{\pm 1}\}$. If $q, s \cdot q \in \mathcal{Y}_{++}$, then*

$$s^{-1} \cdot \Theta_{\kappa_0}(s \cdot q) = \Theta_{\kappa_0}(q).$$

Proof. Strict convexity makes the corresponding cut-and-paste embedded, so only the cut system changes and the marked surface is unchanged. □

Lemma A.6 (Weak walk between consecutive roots)**.** *Let $q \in \mathcal{Y}_{++}$ have logarithmic pair*

$$\begin{pmatrix} c \\ c \end{pmatrix}, \qquad c > 0.$$

Then the weak walk defined by

$$\widetilde{C} = \sigma_1 \sigma_2^{-1} \sigma_1^{-1}$$

preserves the marked surface, and

$$\widetilde{C}^{-1} \cdot \Theta_{\kappa_0}(\widetilde{C} \cdot q) = \Theta_{\kappa_0}(q), \qquad \textit{equivalently} \qquad \Theta_{\kappa_0}(\widetilde{C} \cdot q) = \widetilde{C} \cdot \Theta_{\kappa_0}(q). \tag{72}$$

Proof. The weak walk is

$$\begin{pmatrix} c \\ c \end{pmatrix} \xrightarrow{\sigma_1^{-1}} \begin{pmatrix} c \\ 0 \end{pmatrix} \xrightarrow{\sigma_2^{-1}} \begin{pmatrix} c \\ 0 \end{pmatrix} \xrightarrow{\sigma_1} \begin{pmatrix} c \\ c \end{pmatrix}.$$

The first and last steps are the strict-to-weak and weak-to-strict cut-and-paste moves. It remains to check the middle σ_2^{-1}-step.

Let $\lambda = e^c > 1$. The weak source room has vertices

$$0, \qquad \lambda^{-1} e_1, \qquad \lambda^{-1} e_1 + e_2, \qquad (\lambda^{-1} - 1) e_1 + e_2, \qquad (\lambda^{-1} - 1) e_1.$$

After the σ_2^{-1}-step, the vertices are

$$0, \qquad \lambda^{-1} e_1, \qquad (\lambda^{-1} + 1) e_1 + e_2, \qquad \lambda^{-1} e_1 + e_2, \qquad (\lambda^{-1} - 1) e_1.$$

This is again an embedded weak convex pentagon, and the moved triangle is reattached through the paired side without overlap or fold. Hence the marked surface is unchanged, which gives (72). □

A.5 Proof of the $X_m \neq X_n$

Lemma A.7 (Central rooms at the symmetric parameter)**.** *Put*

$$q_m := Z^m q_0 = [e_1^{(m)}, e_2^{(m)}; \log 2, \log 2].$$

Then

$$e_1^{(m)} = (m+1) - mi, \qquad e_2^{(m)} = m + (1-m)i. \tag{73}$$

The normalized standard room of q_m has vertices

$$\begin{aligned} P_0^{(m)} &= 0, \\ P_1^{(m)} &= \frac{m+1}{2} - \frac{m}{2} i, \\ P_2^{(m)} &= \frac{3m+1}{2} + \frac{2-3m}{2} i, \\ P_3^{(m)} &= \frac{m-1}{2} + \frac{2-m}{2} i, \\ P_4^{(m)} &= -\frac{1}{2} + \frac{1}{2} i. \end{aligned} \tag{74}$$

Its normalized holonomy is

$$\begin{aligned} A_m(z) &= 2z + \frac{m-1}{2} + \frac{2-m}{2}i, \\ B_m(z) &= 2z + \frac{m+3}{2} - \frac{m+2}{2}i, \end{aligned} \tag{75}$$

and

$$\rho_m = T_{-m\delta}\rho_0 T_{m\delta}, \qquad \delta = \frac{1-i}{2}.$$

In particular, the boundary holonomy is translation by the same vector δ for every m.

Proof. For q_0, the door vector is $\delta = (1-i)/2$. Since $\lambda = \mu = 2$, one application of the central room formula adds $2\delta = 1 - i$ to both basis vectors. The door vector is unchanged, because

$$\frac{1}{2}(e_1 + 2\delta) - \frac{1}{2}(e_2 + 2\delta) = \frac{1}{2}(e_1 - e_2).$$

Iteration gives Eq. (73). Substitution in the standard vertex and side-pairing formulas gives Eqs. (74) and (75). Moreover,

$$\omega(e_1^{(m)}, e_2^{(m)}) = 1,$$

so every q_m remains in the positive chamber and its formal room is strict. □

Retain the standard corner words

$$w_0 = 1, \quad w_1 = a^{-1}ba, \quad w_2 = ba, \quad w_3 = a, \quad w_4 = \partial_0^{-1}.$$

Lemma A.8 (Corner labels for every central cut system)**.** *With distinguished corner lift $\widetilde{s}_0$, the corner labels of $Z^{-m}\kappa_0$ are*

$$L_j^{(m)} = \partial_0^{-m} w_j \partial_0^m. \tag{76}$$

Equivalently,

$$\begin{aligned} L_0^{(m)} &= 1, \\ L_1^{(m)} &= \partial_0^{-m} b \partial_0^{m-1}, \\ L_2^{(m)} &= \partial_0^{-m} ab \partial_0^{m-1}, \\ L_3^{(m)} &= \partial_0^{-m} b^{-1} ab \partial_0^{m-1}, \\ L_4^{(m)} &= \partial_0^{-1}. \end{aligned} \tag{77}$$

They develop on X_0 to the vertices in Eq. (74).

Hence a realization of the formal q_m-room on X_0 would contain the side

$$L_1^{(m)}\widetilde{s}_0 \longrightarrow L_2^{(m)}\widetilde{s}_0, \qquad [P_1^{(m)}, P_2^{(m)}]. \tag{78}$$

Proof. Since $j(Z^{-m}) = \mathrm{Inn}(\partial_0^{-m})$, the labels are Eq. (76). The expanded formulas use

$$w_1\partial_0 = b, \qquad w_2\partial_0 = ab, \qquad w_3\partial_0 = b^{-1}ab.$$

Moreover,

$$\widehat{\mathrm{dev}}(L_j^{(m)}\widetilde{s}_0) = \rho_0(w_j)(m\delta) - m\delta = \rho_m(w_j)(0) = P_j^{(m)}.$$

□

Proof. Since $P_2^{(m)} - P_1^{(m)} = m + (1-m)i$, we will show that there is no saddle connection joining $L_1^{(m)}\widetilde{s}_0$ to $L_2^{(m)}\widetilde{s}_0$ whose developed image is

$$[P_1^{(m)}, P_2^{(m)}] = \left\{ z(t) = P_1^{(m)} + t(m + (1-m)i) \;\middle|\; t \in [0,1] \right\} \subset \rho_0(L_1^{(m)})P.$$

Write P for the developed standard room of q_0. We have

$$P = \left\{ x + iy \in \mathbb{C} \;\middle|\; y \geq 0,\ x \leq \frac{1}{2},\ y \leq 1,\ x \geq -\frac{1}{2},\ x + y \geq 0 \right\}.$$

For $g \in \pi_1(S)$, consider the copy $\rho_0(g)P$. When we cross the sides

$$\rho_0(g)(P_0P_1), \qquad \rho_0(g)(P_1P_2), \qquad \rho_0(g)(P_2P_3), \qquad \rho_0(g)(P_3P_4),$$

of $\rho_0(g)P$, we enter another rooms, respectively,

$$\rho_0(ga^{-1})P, \qquad \rho_0(gb)P, \qquad \rho_0(ga)P, \qquad \rho_0(gb^{-1})P.$$

To find all possible local lifts of the segment $[P_1^{(m)}, P_2^{(m)}]$ starting from $L_1^{(m)}\widetilde{s}_0$, we consider a full 3π neighborhood of this singular lift. Its developed image is formed, in order, by the corner sectors of

$$\rho_0(L_1^{(m)})P, \quad \rho_0(L_1^{(m)}a^{-1})P, \quad \rho_0(L_1^{(m)}a^{-1}b^{-1})P, \quad \rho_0(L_1^{(m)}w_1^{-1})P, \quad \rho_0(L_1^{(m)}\partial_0)P.$$

The corresponding angle intervals are

$$[0, 3\pi/4], \quad [-\pi/2, 0], \quad [-\pi, -\pi/2], \quad [-3\pi/2, -\pi], \quad [-9\pi/4, -3\pi/2].$$

The direction of the segment is $m + (1-m)i$. The following table gives its angle and the two sectors where its local lifts are developed to:

m	angle	sectors containing the developed local lifts
$m > 0$	$[-\pi/4, 0]$	$\rho_0(L_1^{(m)}a^{-1})P, \quad \rho_0(L_1^{(m)}\partial_0)P$
$m < 0$	$(\pi/2, 3\pi/4]$	$\rho_0(L_1^{(m)})P, \quad \rho_0(L_1^{(m)}w_1^{-1})P.$

The case $m > 0$.

We now trace the two local lifts. Their developed segments successively enter the developed cells listed below. For each cell, the table gives the first side crossed, the crossing time, and the crossing point in the standard room P. The detailed computations are given in Remarks A.11 and A.12.

cell	exit side	exit time	crossing point in P
$L_1^{(m)}(a^{-1}b)^r a^{-1}P$	P_1P_2	$\dfrac{2r+1}{m}$	$P_1 + \dfrac{2r+1}{m}(P_2 - P_1)$
$L_1^{(m)}(a^{-1}b)^{r+1}P$	P_0P_1	$\dfrac{2r+2}{m-1}$	$P_0 + \dfrac{2r+2}{m-1}(P_1 - P_0)$
$L_1^{(m)}\partial_0(a^{-1}b)^r P$	P_0P_1	$\dfrac{2r+1}{m-1}$	$P_0 + \dfrac{2r+1}{m-1}(P_1 - P_0)$
$L_1^{(m)}\partial_0(a^{-1}b)^r a^{-1}P$	P_1P_2	$\dfrac{2r+2}{m}$	$P_1 + \dfrac{2r+2}{m}(P_2 - P_1).$

(79)

We now use the table to check the endpoints of the two local lifts. We want the local lift to hit a corner at $t = 1$. Thus, when $t = 1$ is a edge crossing time, we first check whether the edge crossing point is a corner. If it is, we then check whether the lift word of this corner gives exactly $L_2^{(m)}\widetilde{s}_0$.

If the corner is given by $L_1^{(m)} E\widetilde{s}_0$, then we verify whether

$$E = (L_1^{(m)})^{-1} L_2^{(m)} = \partial_0^{1-m}(b^{-1}ab)\partial_0^{m-1}.$$

We shall distinguish these relative endpoint words using abelianization: since $[\partial_0]_{\mathrm{ab}} = 0$, we have $\left[(L_1^{(m)})^{-1} L_2^{(m)}\right]_{\mathrm{ab}} = (1, 0)$.

If $m = 2k+1$, the second local lift has no corner crossing at $t = 1$, since its crossing times

$$\frac{2r+1}{m-1} = \frac{2r+1}{2k} \qquad \text{and} \qquad \frac{2r+2}{m} = \frac{2r+2}{2k+1}$$

cannot be 1 .

For the first local lift, the crossing time $\frac{2r+1}{m}$ equals 1 exactly when $r = k$. At this time, the crossing point is the P_2-corner. Hence the corresponding singular lift is

$$L_1^{(m)}(a^{-1}b)^k a^{-1} w_2 \widetilde{s}_0 = L_1^{(m)}(a^{-1}b)^k w_1 \widetilde{s}_0.$$

But

$$[(a^{-1}b)^k w_1]_{\mathrm{ab}} = (-k, k+1) \neq (1, 0).$$

Thus this corner is a singular lift, but it is not the prescribed terminal lift $L_2^{(m)}\widetilde{s}_0$.

If $m = 2k$, the first local lift has no corner crossing at $t = 1$, since its crossing times

$$\frac{2r+1}{m} = \frac{2r+1}{2k} \qquad \text{and} \qquad \frac{2r+2}{m-1} = \frac{2r+2}{2k-1}$$

cannot be 1.

For the second local lift, the crossing time $\frac{2r+2}{m}$ equals 1 exactly when $r = k-1$. At this time, the crossing point is the P_2-corner. Hence the corresponding singular lift is

$$L_1^{(m)}\partial_0(a^{-1}b)^{k-1}a^{-1}w_2\widetilde{s}_0 = L_1^{(m)}\partial_0(a^{-1}b)^{k-1}w_1\widetilde{s}_0.$$

But

$$[\partial_0(a^{-1}b)^{k-1}w_1]_{\mathrm{ab}} = (1-k, k) \neq (1, 0).$$

Thus this corner is a singular lift, but it is not the prescribed terminal lift $L_2^{(m)}\widetilde{s}_0$.

The case $m < 0$.

We now trace the two local lifts. Their developed segments successively enter the developed cells listed below. For each cell, the table gives the first side crossed, the crossing time, and the crossing point in the standard room P. The computations are given in Remarks A.11 and A.12.

cell	exit side	exit time	crossing point in P
$L_1^{(m)}(b^{-1}a)^r P$	P_3P_4	$\dfrac{2r+1}{-m}$	$P_4 + \dfrac{2r+1}{-m}(P_3 - P_4)$
$L_1^{(m)}(b^{-1}a)^r b^{-1} P$	P_2P_3	$\dfrac{2r+2}{1-m}$	$P_3 + \dfrac{2r+2}{1-m}(P_2 - P_3)$
$L_1^{(m)} w_1^{-1}(ab^{-1})^r P$	P_2P_3	$\dfrac{2r+1}{1-m}$	$P_3 + \dfrac{2r+1}{1-m}(P_2 - P_3)$
$L_1^{(m)} w_1^{-1}(ab^{-1})^r a P$	P_3P_4	$\dfrac{2r+2}{-m}$	$P_4 + \dfrac{2r+2}{-m}(P_3 - P_4)$.

(80)

We now use the table to check the endpoints of the two local lifts. We want the local lift to hit a corner at $t = 1$. Thus, when $t = 1$ is an edge crossing time, we first check whether the edge crossing point is a corner. If it is, we then check whether the lift word of this corner gives exactly $L_2^{(m)}\widetilde{s}_0$. If the corner is given by $L_1^{(m)} E\widetilde{s}_0$, then we verify whether

$$E = (L_1^{(m)})^{-1} L_2^{(m)} = \partial_0^{1-m}(b^{-1}ab)\partial_0^{m-1}.$$

We shall distinguish these relative endpoint words using abelianization: since $[\partial_0]_{\mathrm{ab}} = 0$, we have $\left[(L_1^{(m)})^{-1}L_2^{(m)}\right]_{\mathrm{ab}} = (1,0)$.

If $-m = 2k$, the first local lift has no corner crossing at $t = 1$, since its crossing times

$$\frac{2r+1}{-m} = \frac{2r+1}{2k} \qquad \text{and} \qquad \frac{2r+2}{1-m} = \frac{2r+2}{2k+1}$$

cannot be 1.

For the second local lift, the crossing time $\frac{2r+2}{-m}$ equals 1 exactly when $r = k-1$. At this time, the crossing point is the P_3-corner. Hence the corresponding singular lift is

$$L_1^{(m)} w_1^{-1}(ab^{-1})^{k-1} a w_3 \widetilde{s}_0 = L_1^{(m)} w_1^{-1}(ab^{-1})^{k-1} a^2 \widetilde{s}_0.$$

But

$$\left[w_1^{-1}(ab^{-1})^{k-1}a^2\right]_{\mathrm{ab}} = (k+1,-k) \neq (1,0).$$

Thus this corner is a singular lift, but it is not the prescribed terminal lift $L_2^{(m)}\widetilde{s}_0$.

If $-m = 2k+1$, the second local lift has no corner crossing at $t = 1$, since its crossing times

$$\frac{2r+1}{1-m} = \frac{2r+1}{2k+2} \qquad \text{and} \qquad \frac{2r+2}{-m} = \frac{2r+2}{2k+1}$$

cannot be 1.

For the first local lift, the crossing time $\frac{2r+1}{-m}$ equals 1 exactly when $r = k$. At this time, the crossing point is the P_3-corner. Hence the corresponding singular lift is

$$L_1^{(m)}(b^{-1}a)^k w_3 \widetilde{s}_0 = L_1^{(m)}(b^{-1}a)^k a \widetilde{s}_0.$$

But

$$\left[(b^{-1}a)^k a\right]_{\mathrm{ab}} = (k+1,-k).$$

For $k \geq 1$, this is different from $(1,0)$. Thus this corner is a singular lift, but it is not the prescribed terminal lift $L_2^{(m)}\widetilde{s}_0$.

It remains to consider $k = 0$, namely $m = -1$. The corner word is a, while the prescribed relative endpoint word is $(L_1^{(-1)})^{-1}L_2^{(-1)} = \partial_0^2 b^{-1}ab\partial_0^{-2} = (a^{-1}b^{-1}ab)^2 a (b^{-1}a^{-1}ba) \neq a$. □

We shall also use the following formulas in the computations of the remarks below:

$$\begin{aligned} \rho_0(a^{-1}b) &= T_{1-i}, & \rho_0(\partial_0) &= T_{(1-i)/2}, \\ \rho_0(b^{-1}a) &= T_{-1+i}, & \rho_0(ab^{-1}) &= T_{-2+2i}, \\ \rho_0(w_1)(z) &= 2z + \frac{1}{2}, & \rho_0(L_1^{(m)})(z) &= 2z + P_1^{(m)}. \end{aligned}$$

Remark A.9 (Computation of the first positive trace). We start in the developed cell

$$\rho_0\left(L_1^{(m)}a^{-1}\right)P.$$

Pulling the segment back to P, we obtain

$$u(t) = \rho_0\left(L_1^{(m)}a^{-1}\right)^{-1}\left(P_1^{(m)} + t(m + (1-m)i)\right) = -\frac{1}{2} + mt + i(1 + (1-m)t).$$

Since P_1P_2 is given by $\Re z = \frac{1}{2}$, the segment crosses P_1P_2 at $t = \frac{1}{m}$, at the point

$$u\left(\frac{1}{m}\right) = P_1 + \frac{1}{m}(P_2 - P_1).$$

Before this time, the other four inequalities of P are not violated. Hence the segment crosses P_1P_2 and enters $\rho_0\left(L_1^{(m)}(a^{-1}b)\right) P$.

In this new room, pulling the same segment back to P gives

$$u(t) = \rho_0\left(L_1^{(m)}(a^{-1}b)\right)^{-1}\left(P_1^{(m)} + t(m + (1-m)i)\right) = -1 + \frac{mt}{2} + i\left(1 + \frac{(1-m)t}{2}\right).$$

Since P_0P_1 is given by $\Im z = 0$, the segment crosses P_0P_1 at $t = \frac{2}{m-1}$, at the point

$$u\left(\frac{2}{m-1}\right) = P_0 + \frac{2}{m-1}(P_1 - P_0).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_0P_1 and enters $\rho_0\left(L_1^{(m)}(a^{-1}b)a^{-1}\right) P$.

Thus the first two crossings are

$$\rho_0\left(L_1^{(m)}a^{-1}\right) P \xrightarrow{P_1P_2} \rho_0\left(L_1^{(m)}(a^{-1}b)\right) P \xrightarrow{P_0P_1} \rho_0\left(L_1^{(m)}(a^{-1}b)a^{-1}\right) P.$$

Now the last room has exactly the same form as the first one, with one extra factor $a^{-1}b$. Repeating the same calculation inductively, suppose the current developed cell is

$$\rho_0\left(L_1^{(m)}(a^{-1}b)^r a^{-1}\right) P.$$

To see where the segment leaves this cell, we pull it back to the standard room P:

$$\begin{aligned} u(t) &= \rho_0\left(L_1^{(m)}(a^{-1}b)^r a^{-1}\right)^{-1}\left(P_1^{(m)} + t(m + (1-m)i)\right) \\ &= -\left(2r + \frac{1}{2}\right) + mt + i\left(2r + 1 + (1-m)t\right). \end{aligned}$$

Since the side P_1P_2 of P is given by $\Re z = \frac{1}{2}$, the segment crosses P_1P_2 when $-\left(2r+\frac{1}{2}\right) + mt = \frac{1}{2}$, which gives $t = \frac{2r+1}{m}$. Since $u\left(\frac{2r+1}{m}\right) = \frac{1}{2} + \frac{2r+1}{m}i$, with $P_1 = \frac{1}{2}$ and $P_2 = \frac{1}{2} + i$, the crossing point is

$$P_1 + \frac{2r+1}{m}(P_2 - P_1)$$

. Before this time, the other four inequalities of P are not violated, so the segment crosses P_1P_2 and enters the next room. Since crossing P_1P_2 sends $\rho_0(g)P$ to $\rho_0(gb)P$, we have $\rho_0\left(L_1^{(m)}(a^{-1}b)^r a^{-1}\right) P \longrightarrow \rho_0\left(L_1^{(m)}(a^{-1}b)^{r+1}\right) P$.

We continue with the same segment and pull it back from $\rho_0\left(L_1^{(m)}(a^{-1}b)^{r+1}\right) P$ to the standard room P:

$$\begin{aligned} u(t) &= \rho_0\left(L_1^{(m)}(a^{-1}b)^{r+1}\right)^{-1}\left(P_1^{(m)} + t(m + (1-m)i)\right) \\ &= -(r+1) + \frac{mt}{2} + i\left(r + 1 + \frac{(1-m)t}{2}\right). \end{aligned}$$

Since the side P_0P_1 of P is given by $\Im z = 0$, the segment crosses P_0P_1 when $r + 1 + \frac{(1-m)t}{2} = 0$, which gives $t = \frac{2r+2}{m-1}$. At this time, $u\left(\frac{2r+2}{m-1}\right) = \frac{r+1}{m-1} = \frac{t}{2}$. Since $P_0 = 0$ and $P_1 = \frac{1}{2}$, the crossing point on P_0P_1 is

$$P_0 + t(P_1 - P_0).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_0P_1 and enters the next room. Since crossing P_0P_1 sends $\rho_0(g)P$ to $\rho_0(ga^{-1})P$, we have $\rho_0\left(L_1^{(m)}(a^{-1}b)^{r+1}\right) P \longrightarrow \rho_0\left(L_1^{(m)}(a^{-1}b)^{r+1}a^{-1}\right) P$.

Therefore one complete step is $\rho_0\left(L_1^{(m)}(a^{-1}b)^r a^{-1}\right) P \xrightarrow{P_1P_2} \rho_0\left(L_1^{(m)}(a^{-1}b)^{r+1}\right) P \xrightarrow{P_0P_1}$ $\rho_0\left(L_1^{(m)}(a^{-1}b)^{r+1}a^{-1}\right) P$, and replacing r by $r+1$ repeats the same calculation.

Remark A.10 (Computation of the second positive trace). We start the second local lift in the developed cell

$$\rho_0\Big(L_1^{(m)}\partial_0\Big)\,P.$$

Pulling the segment back to P, we obtain

$$u(t)=\rho_0\Big(L_1^{(m)}\partial_0\Big)^{-1}\Big(P_1^{(m)}+t(m+(1-m)i)\Big)=-\frac{1}{2}+\frac{mt}{2}+i\left(\frac{1}{2}+\frac{(1-m)t}{2}\right).$$

Since P_0P_1 is given by $\Im z=0$, the segment crosses P_0P_1 at $t=\frac{1}{m-1}$, at the point

$$u\left(\frac{1}{m-1}\right)=P_0+\frac{1}{m-1}(P_1-P_0).$$

Before this time, the other four inequalities of P are not violated. Hence the segment crosses P_0P_1 and enters $\rho_0\Big(L_1^{(m)}\partial_0 a^{-1}\Big)\,P$.

In this new room, pulling the same segment back to P gives

$$u(t)=\rho_0\Big(L_1^{(m)}\partial_0 a^{-1}\Big)^{-1}\Big(P_1^{(m)}+t(m+(1-m)i)\Big)=-\frac{3}{2}+mt+i(2+(1-m)t).$$

Since P_1P_2 is given by $\Re z=\frac{1}{2}$, the segment crosses P_1P_2 at $t=\frac{2}{m}$, at the point

$$u\left(\frac{2}{m}\right)=P_1+\frac{2}{m}(P_2-P_1).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_1P_2 and enters $\rho_0\Big(L_1^{(m)}\partial_0(a^{-1}b)\Big)\,P$.

Thus the first two crossings are

$$\rho_0\Big(L_1^{(m)}\partial_0\Big)\,P\xrightarrow{P_0P_1}\rho_0\Big(L_1^{(m)}\partial_0 a^{-1}\Big)\,P\xrightarrow{P_1P_2}\rho_0\Big(L_1^{(m)}\partial_0(a^{-1}b)\Big)\,P.$$

Now the last room has exactly the same form as the first one, with one extra factor $a^{-1}b$. Repeating the same calculation inductively, suppose the current developed cell is

$$\rho_0\Big(L_1^{(m)}\partial_0(a^{-1}b)^r\Big)\,P.$$

Pulling the segment back to P, we obtain

$$\begin{aligned}u(t)&=\rho_0\Big(L_1^{(m)}\partial_0(a^{-1}b)^r\Big)^{-1}\Big(P_1^{(m)}+t(m+(1-m)i)\Big)\\&=-\frac{2r+1}{2}+\frac{mt}{2}+i\left(\frac{2r+1}{2}+\frac{(1-m)t}{2}\right).\end{aligned}$$

Since P_0P_1 is given by $\Im z=0$, the segment crosses P_0P_1 at $t=\frac{2r+1}{m-1}$. At this time,

$$u\left(\frac{2r+1}{m-1}\right)=P_0+\frac{2r+1}{m-1}(P_1-P_0).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_0P_1 and enters $\rho_0\Big(L_1^{(m)}\partial_0(a^{-1}b)^r a^{-1}\Big)\,P$.

We continue with the same segment and pull it back from $\rho_0\Big(L_1^{(m)}\partial_0(a^{-1}b)^r a^{-1}\Big)\,P$ to the standard room P:

$$\begin{aligned}u(t)&=\rho_0\Big(L_1^{(m)}\partial_0(a^{-1}b)^r a^{-1}\Big)^{-1}\Big(P_1^{(m)}+t(m+(1-m)i)\Big)\\&=-\left(2r+\frac{3}{2}\right)+mt+i\left(2r+2+(1-m)t\right).\end{aligned}$$

Since P_1P_2 is given by $\Re z = \frac{1}{2}$, the segment crosses P_1P_2 at $t = \frac{2r+2}{m}$. At this time,

$$u\left(\frac{2r+2}{m}\right) = P_1 + \frac{2r+2}{m}(P_2 - P_1).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_1P_2 and enters $\rho_0\left(L_1^{(m)}\partial_0(a^{-1}b)^{r+1}\right) P$.

Therefore one complete step is $\rho_0\left(L_1^{(m)}\partial_0(a^{-1}b)^{r}\right) P \xrightarrow{P_0P_1} \rho_0\left(L_1^{(m)}\partial_0(a^{-1}b)^{r}a^{-1}\right) P \xrightarrow{P_1P_2}$ $\rho_0\left(L_1^{(m)}\partial_0(a^{-1}b)^{r+1}\right) P$, and replacing r by $r+1$ repeats the same calculation.

Remark A.11 (Computation of the first negative trace). We start in the developed cell

$$\rho_0\left(L_1^{(m)}\right) P.$$

Pulling the segment back to P, we obtain

$$u(t) = \rho_0\left(L_1^{(m)}\right)^{-1}\left(P_1^{(m)} + t(m + (1-m)i)\right) = \frac{mt}{2} + i\frac{(1-m)t}{2}.$$

Since P_3P_4 is given by $\Re z = -\frac{1}{2}$, the segment crosses P_3P_4 at $t = \frac{1}{-m}$, at the point

$$u\left(\frac{1}{-m}\right) = P_4 + \frac{1}{-m}(P_3 - P_4).$$

Before this time, the other four inequalities of P are not violated. Hence the segment crosses P_3P_4 and enters $\rho_0\left(L_1^{(m)}b^{-1}\right) P$.

In this new room, pulling the same segment back to P gives

$$u(t) = \rho_0\left(L_1^{(m)}b^{-1}\right)^{-1}\left(P_1^{(m)} + t(m + (1-m)i)\right) = \frac{3}{2} + mt + i(-1 + (1-m)t).$$

Since P_2P_3 is given by $\Im z = 1$, the segment crosses P_2P_3 at $t = \frac{2}{1-m}$, at the point

$$u\left(\frac{2}{1-m}\right) = P_3 + \frac{2}{1-m}(P_2 - P_3).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_2P_3 and enters $\rho_0\left(L_1^{(m)}b^{-1}a\right) P = \rho_0\left(L_1^{(m)}(b^{-1}a)\right) P$.

Thus the first two crossings are

$$\rho_0\left(L_1^{(m)}\right) P \xrightarrow{P_3P_4} \rho_0\left(L_1^{(m)}b^{-1}\right) P \xrightarrow{P_2P_3} \rho_0\left(L_1^{(m)}(b^{-1}a)\right) P.$$

Now the last room has exactly the same form as the first one, with one extra factor $b^{-1}a$. Repeating the same calculation inductively, suppose the current developed cell is

$$\rho_0\left(L_1^{(m)}(b^{-1}a)^r\right) P.$$

Pulling the segment back to P, we obtain

$$\begin{aligned} u(t) &= \rho_0\left(L_1^{(m)}(b^{-1}a)^r\right)^{-1}\left(P_1^{(m)} + t(m + (1-m)i)\right) \\ &= r + \frac{mt}{2} + i\left(-r + \frac{(1-m)t}{2}\right). \end{aligned}$$

Since P_3P_4 is given by $\Re z = -\frac{1}{2}$, the segment crosses P_3P_4 at $t = \frac{2r+1}{-m}$. At this time,

$$u\left(\frac{2r+1}{-m}\right) = P_4 + \frac{2r+1}{-m}(P_3 - P_4).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_3P_4 and enters $\rho_0\Big(L_1^{(m)}(b^{-1}a)^r b^{-1}\Big)\,P$.

We continue with the same segment and pull it back from $\rho_0\Big(L_1^{(m)}(b^{-1}a)^r b^{-1}\Big)\,P$ to the standard room P:

$$\begin{aligned} u(t) &= \rho_0\Big(L_1^{(m)}(b^{-1}a)^r b^{-1}\Big)^{-1}\left(P_1^{(m)}+t(m+(1-m)i)\right) \\ &= 2r+\frac{3}{2}+mt+i\left(-2r-1+(1-m)t\right). \end{aligned}$$

Since P_2P_3 is given by $\Im z=1$, the segment crosses P_2P_3 at $t=\frac{2r+2}{1-m}$. At this time,

$$u\left(\frac{2r+2}{1-m}\right)=P_3+\frac{2r+2}{1-m}(P_2-P_3).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_2P_3 and enters $\rho_0\Big(L_1^{(m)}(b^{-1}a)^{r+1}\Big)\,P$.

Therefore one complete step is $\rho_0\Big(L_1^{(m)}(b^{-1}a)^r\Big)\,P \xrightarrow{P_3P_4} \rho_0\Big(L_1^{(m)}(b^{-1}a)^r b^{-1}\Big)\,P \xrightarrow{P_2P_3} \rho_0\Big(L_1^{(m)}(b^{-1}a)^{r+1}\Big)\,P$, and replacing r by $r+1$ repeats the same calculation.

Remark A.12 (Computation of the second negative trace)*.* We start the second local lift in the developed cell

$$\rho_0\Big(L_1^{(m)}w_1^{-1}\Big)\,P.$$

Pulling the segment back to P, we obtain

$$u(t)=\rho_0\Big(L_1^{(m)}w_1^{-1}\Big)^{-1}\left(P_1^{(m)}+t(m+(1-m)i)\right)=\frac{1}{2}+mt+i(1-m)t.$$

Since P_2P_3 is given by $\Im z=1$, the segment crosses P_2P_3 at $t=\frac{1}{1-m}$, at the point

$$u\left(\frac{1}{1-m}\right)=P_3+\frac{1}{1-m}(P_2-P_3).$$

Before this time, the other four inequalities of P are not violated. Hence the segment crosses P_2P_3 and enters $\rho_0\Big(L_1^{(m)}w_1^{-1}a\Big)\,P$.

In this new room, pulling the same segment back to P gives

$$u(t)=\rho_0\Big(L_1^{(m)}w_1^{-1}a\Big)^{-1}\left(P_1^{(m)}+t(m+(1-m)i)\right)=\frac{1}{2}+\frac{mt}{2}+i\left(-\frac{1}{2}+\frac{(1-m)t}{2}\right).$$

Since P_3P_4 is given by $\Re z=-\frac{1}{2}$, the segment crosses P_3P_4 at $t=\frac{2}{-m}$, at the point

$$u\left(\frac{2}{-m}\right)=P_4+\frac{2}{-m}(P_3-P_4).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_3P_4 and enters $\rho_0\Big(L_1^{(m)}w_1^{-1}ab^{-1}\Big)\,P=\rho_0\Big(L_1^{(m)}w_1^{-1}(ab^{-1})\Big)\,P$.

Thus the first two crossings are

$$\rho_0\Big(L_1^{(m)}w_1^{-1}\Big)\,P \xrightarrow{P_2P_3} \rho_0\Big(L_1^{(m)}w_1^{-1}a\Big)\,P \xrightarrow{P_3P_4} \rho_0\Big(L_1^{(m)}w_1^{-1}(ab^{-1})\Big)\,P.$$

Now the last room has exactly the same form as the first one, with one extra factor ab^{-1}. Repeating the same calculation inductively, suppose the current developed cell is

$$\rho_0\Big(L_1^{(m)}w_1^{-1}(ab^{-1})^r\Big)\,P.$$

Pulling the segment back to P, we obtain

$$\begin{aligned}u(t) &= \rho_0\left(L_1^{(m)} w_1^{-1}(ab^{-1})^r\right)^{-1}\left(P_1^{(m)} + t(m + (1-m)i)\right)\\ &= 2r + \frac{1}{2} + mt + i\left(-2r + (1-m)t\right).\end{aligned}$$

Since P_2P_3 is given by $\Im z = 1$, the segment crosses P_2P_3 at $t = \frac{2r+1}{1-m}$. At this time,

$$u\left(\frac{2r+1}{1-m}\right) = P_3 + \frac{2r+1}{1-m}(P_2 - P_3).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_2P_3 and enters $\rho_0\left(L_1^{(m)} w_1^{-1}(ab^{-1})^r a\right) P$.

We continue with the same segment and pull it back from $\rho_0\left(L_1^{(m)} w_1^{-1}(ab^{-1})^r a\right) P$ to the standard room P:

$$\begin{aligned}u(t) &= \rho_0\left(L_1^{(m)} w_1^{-1}(ab^{-1})^r a\right)^{-1}\left(P_1^{(m)} + t(m + (1-m)i)\right)\\ &= r + \frac{1}{2} + \frac{mt}{2} + i\left(-r - \frac{1}{2} + \frac{(1-m)t}{2}\right).\end{aligned}$$

Since P_3P_4 is given by $\Re z = -\frac{1}{2}$, the segment crosses P_3P_4 at $t = \frac{2r+2}{-m}$. At this time,

$$u\left(\frac{2r+2}{-m}\right) = P_4 + \frac{2r+2}{-m}(P_3 - P_4).$$

Before this time, the other four inequalities of P are not violated, so the segment crosses P_3P_4 and enters $\rho_0\left(L_1^{(m)} w_1^{-1}(ab^{-1})^{r+1}\right) P$.

Therefore one complete step is $\rho_0\left(L_1^{(m)} w_1^{-1}(ab^{-1})^r\right) P \xrightarrow{P_2P_3} \rho_0\left(L_1^{(m)} w_1^{-1}(ab^{-1})^r a\right) P \xrightarrow{P_3P_4} \rho_0\left(L_1^{(m)} w_1^{-1}(ab^{-1})^{r+1}\right) P$, and replacing r by $r+1$ repeats the same calculation.

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